A correlated prior on galaxy bias across thin redshift bins

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1 Introduction: why an intermediate bias model

A Fisher forecast of galaxy clustering in thin redshift bins must decide how to parametrize the galaxy bias of each bin. The bins belong to a few broad groups: the DESI tracer samples and their chunks, such as BGS, LRG1, LRG2, LRG3, ELG1 and ELG2. Within one group the galaxies are selected the same way, so their biases are expected to be similar but not identical.

The default model ties all bins of a group to a single amplitude. If $D(z)$ is the linear growth factor, normalized to $D(0) = 1$, the tied model sets

$$b_i \;=\; \frac{B_g}{D(z_i)}, \tag{1}$$

where b_i is the linear bias of bin i, z_i is the bin's redshift, and B_g is one free amplitude per group g. The $1/D$ evolution keeps the clustering amplitude $b(z)\,D(z)$ of the group constant, the behavior measured for DESI-like luminous red galaxies [6]. It is not the law of a fixed set of galaxies that follow the flow of matter ("passive evolution"); Section 3.1 derives that law and the small difference between the two.

The opposite extreme frees one bias per bin, with flat priors and no connection between neighbors ("independent"). The dark-energy figure of merit of the forecast depends on which extreme is chosen, and the question is how much of it rests on the tying itself. To answer that, this note builds the intermediate model: bins may deviate from the group's trend, but neighboring bins must deviate *together*.

2 The models

2.1 Tied, independent, and correlated biases

The model multiplies the tied bias by a fractional deviation δ_i,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1+\delta_i\big), \tag{2}$$

with B_g still free (flat prior). The deviations receive a Gaussian prior that is correlated inside a group,

$$\big\langle \delta_i\,\delta_j \big\rangle \;=\; \sigma_b^2\,\exp\!\left(-\,\frac{|z_i - z_j|}{L}\right), \tag{3}$$

and zero across groups. Here σ_b is the prior standard deviation of a single bin's fractional deviation, and L is the correlation length in redshift: bins much closer than L deviate almost as one, bins much farther apart deviate independently.

In the Fisher matrix the convenient parameters are the per-bin amplitudes

$$\beta_i \;\equiv\; B_g\,\big(1+\delta_i\big), \qquad \text{fiducial } \beta_i = B_g, \tag{4}$$

so that $b_i = \beta_i / D(z_i)$. Since $\beta_i - B_g = B_g\,\delta_i$, the prior covariance of the β_i within one group is

$$\mathsf{C} \;=\; s^2\,\mathsf{R}, \qquad s \;=\; \sigma_b\,B_g, \qquad \mathsf{R}_{ij} \;=\; \exp\!\left(-\,\frac{|z_i - z_j|}{L}\right). \tag{5}$$

The spread s carries the units of a bias amplitude; the correlation matrix R is dimensionless, with ones on the diagonal.

2.2 What σ_b and L mean

The kernel of Eq. (3) has two parameters, and they control two different things. The diagonal gives the wander of a single bin,

$$\mathrm{Var}(\delta_i) \;=\; \sigma_b^2, \tag{6}$$

so σ_b sets how far any one bin may drift from the group's trend $B_g/D(z)$. The correlation coefficient r of two bins i and j follows from the same kernel. By definition,

$$r \;=\; \frac{\langle \delta_i \delta_j \rangle}{\sqrt{\langle \delta_i^2 \rangle\,\langle \delta_j^2 \rangle}} \;=\; \frac{\sigma_b^2\, e^{-\Delta z/L}}{\sigma_b^2} \;=\; e^{-\Delta z/L}, \tag{7}$$

with $\Delta z = |z_i - z_j|$ the bin separation. The amplitude σ_b cancels: the correlation is set by the separation and L alone.

The jump between two bins now follows from the two pieces above: with $\mathrm{Var}(\delta_i) = \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \sigma_b^2\, e^{-\Delta z/L}$, the variance of the difference is $\mathrm{Var}(\delta_i) + \mathrm{Var}(\delta_j) - 2\,\mathrm{Cov}(\delta_i, \delta_j)$:

$$\mathrm{Var}(\delta_i - \delta_j) \;=\; 2\,\sigma_b^2\left(1 - e^{-\Delta z/L}\right). \tag{8}$$

For separations much smaller than the correlation length, $\Delta z \ll L$, the exponential expands as $e^{-x} \approx 1 - x$, and

$$\mathrm{Var}(\delta_i - \delta_j) \;\approx\; \frac{2\,\sigma_b^2\,\Delta z}{L}, \qquad \text{rms jump} \;\approx\; \sigma_b\sqrt{\frac{2\,\Delta z}{L}}. \tag{9}$$

So L sets the correlation between bins, σ_b sets the wander of each bin, and *together* they set the jump a thin bin may take relative to its neighbor.

The combination that controls small separations is the roughness

$$\kappa \;\equiv\; \frac{\sigma_b}{\sqrt{L}}, \qquad \mathrm{Var}(\delta_i - \delta_j) \;\approx\; 2\,\kappa^2\,\Delta z. \tag{10}$$

The second form is the signature of diffusion: the variance of the increment grows linearly with the separation, exactly as the distance walked by a random walker grows linearly in time. On scales well below L the deviation curve $\delta(z)$ therefore looks like a random walk with diffusion coefficient κ^2; only on scales beyond L does the finite wander σ_b pull it back toward zero.

Worked example. Take neighboring thin bins with $\Delta z = 0.02$ and correlation length $L = 0.05$. Then $\Delta z/L = 0.4$ and

$$r \;=\; e^{-0.4} \;=\; 0.670. \tag{11}$$

The exact rms jump from Eq. (8) is $\sigma_b\sqrt{2\,(1-0.670)} = 0.81\,\sigma_b$: for $\sigma_b = 10\%$ the neighboring bins may differ by 8.1% (one sigma), and for $\sigma_b = 1\%$ by 0.81%. The Taylor form of Eq. (9) gives instead $\sigma_b\sqrt{0.8} \;=\; 0.89\,\sigma_b$, an 8.9% jump. The approximation overestimates because $\Delta z/L = 0.4$ is not actually small: the expansion replaces $1 - e^{-x}$ by x, and x exceeds $1 - e^{-x}$ already at second order ($x^2/2$), by about twenty percent here; the square root halves that, so the rms jump is overestimated by about ten percent.

2.3 The free group amplitude and the projection

A plain Gaussian prior with covariance C would penalize moving all β_i of a group up or down together, but that uniform shift is exactly the free amplitude B_g, which the tied model never constrains. Keeping the penalty would make every comparison against the tied model unfair. The repair removes the uniform direction from the prior: with $\mathbf{1} = (1, \ldots, 1)^{\mathsf{T}}$ the ones vector with n entries (n the number of bins in the group), the prior precision of a group's amplitudes is the projection

$$\mathsf{P} \;=\; \mathsf{C}^{-1} \;-\; \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{12}$$

Four statements, proven in the appendices, carry the construction. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing, so B_g stays as free as in the tied model. A pure shape change keeps exactly the penalty the plain prior gives it. P is positive semidefinite with rank $n - 1$: of a group's n directions, one is free and the other $n - 1$ are charged. And P is precisely the prior obtained by sampling B_g with a flat prior and marginalizing it. Appendix A computes P for two bins, Appendix B derives these statements three ways, and Appendix C treats n bins, including the tied and independent limits.

Across groups the prior of Eq. (3) is zero, so the full prior precision is block diagonal: one projected block per group, each group with its own free amplitude. How the blocks enter the Fisher matrix is Appendix H.

3 Which L and σ_b are physical

Which deviations from the tied trend does known galaxy-bias physics produce? One case of bias evolution can be solved in closed form: a population whose members are conserved. This section derives that case, measures its deviation from the tied trend, compares the deviation with what the prior permits, weighs the published measurements, and ends at the scanned grid and its plausible region.

3.1 A conserved population: the bias continuity equation

Take a sample whose membership never changes: no galaxy enters, none leaves. Assume the galaxies move with the matter on large scales (no velocity bias), and work in linear theory.

Under these assumptions the matter overdensity δ_m (the fractional fluctuation of the matter density) and the galaxy-count overdensity δ_g obey the same linear continuity equation, with the same peculiar velocity field on the right-hand side: number conservation gives each left-hand

side, and the common motion gives the shared right-hand side. Subtracting the two equations, the excess clumping $\delta_g - \delta_m$ is frozen: whatever clumping difference the sample is born with never changes. Appendix F spells out every step.

Let z_c be a reference redshift where the sample has linear bias b_c (linear bias means $\delta_g = b\,\delta_m$), and let the matter grow as $\delta_m \propto D$. Carrying the frozen difference from z_c to any other redshift gives [1]

$$b(z) - 1 \;=\; \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}. \tag{13}$$

The excess clumping set when the galaxies formed stays fixed while the matter clustering grows underneath it, so the bias decays toward one with time. Tegmark and Peebles [2] extend the calculation to stochastic bias, where the conserved galaxy field traces the matter imperfectly (correlation $\rho_{gm} < 1$), and find the same decay of b toward one; Appendix F summarizes their result.

The tied shape $b = B/D$ freezes a different quantity. Constant $b\,D$ means $\delta_g = b\,\delta_m \propto (B/D)\,D = B$: the galaxy clustering amplitude itself does not grow. A conserved population never does that; its δ_g keeps growing with the matter piece $\delta_m(z)$ (Appendix F). Constant $b\,D$ instead describes a selection: at each redshift it picks a different and rarer set of objects, whose stronger clumping compensates the smaller D.

3.2 The deviation from the tied trend

Place the reference at a group's midpoint, z_c, with $b_c = B/D(z_c)$ from the group's fiducial amplitude B. The tied bias is $b_{\rm tied}(z) = b_c\,D(z_c)/D(z)$ and the conserved bias is Eq. (13). Their ratio defines the deviation $\delta(z)$ of Eq. (2), $b = b_{\rm tied}(1+\delta)$:

$$1 + \delta(z) \;=\; \frac{1 + (b_c - 1)\,D(z_c)/D(z)}{b_c\,D(z_c)/D(z)} \;=\; \frac{D(z)/D(z_c) + b_c - 1}{b_c}, \qquad \delta(z) \;=\; \frac{D(z)/D(z_c) - 1}{b_c}. \tag{14}$$

At the midpoint $\delta = 0$. At lower redshift the matter has grown (D larger than $D(z_c)$), so $\delta > 0$; at higher redshift $\delta < 0$. Because $D(z)$ is smooth, $\delta(z)$ is nearly linear in z across a group.

Table 1 evaluates Eq. (14) in flat ΛCDM with $\Omega_m = 0.3$, using the project's fiducial amplitudes. In the language of the prior, a conserved population is a smooth tilt: coherent over the whole group, so L longer than the group width, with an amplitude of a few percent at the group ends and a step of 0.3–0.7% between neighbors at $\Delta z = 0.02$.

Group	B	z range	b_c	δ at the group ends	per $\Delta z = 0.05$
BGS	1.34	0.1–0.4	1.5	+5.3%, −4.9%	1.7%
LRG1	1.70	0.4–0.6	2.2	+2.4%, −2.2%	1.1%
LRG3	1.70	0.8–1.1	2.7	+2.6%, −2.3%	0.8%

Table 1: The conserved-population deviation $\delta(z)$ of Eq. (14) from the tied trend, for flat ΛCDM with $\Omega_m = 0.3$, the reference z_c at the group midpoint, and $b_c = B/D(z_c)$ from the project fiducials. The last column is the change of δ across one $\Delta z = 0.05$ step at the midpoint.

Figure 1 draws the shapes behind the table. Look at any group: the dashed conserved curve is flatter than the solid B_g/D, because the frozen excess clumping is divided by a growing D,

so the conserved bias rises more slowly toward high redshift. On BGS, the dash-dot flux-limit curve is the opposite: much steeper than B_g/D, and convex.

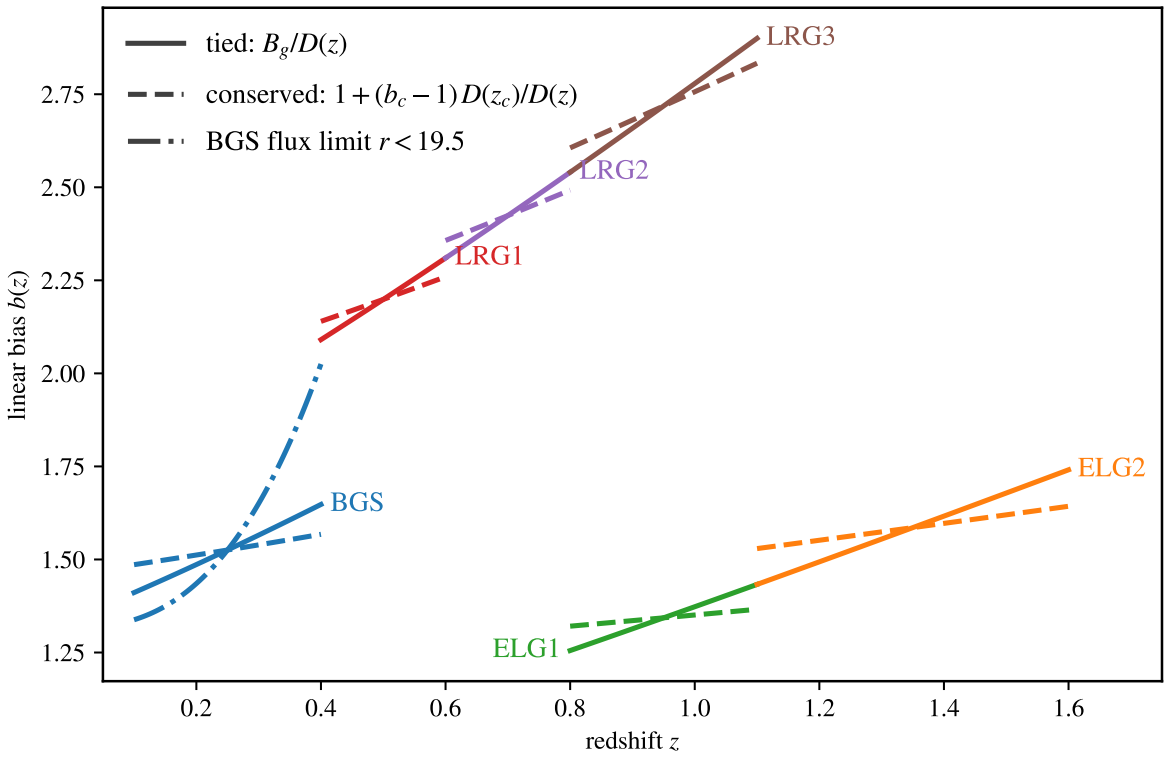


Figure 1: The bias $b(z)$ of every group. Solid: the tied shape $B_g/D(z)$. Dashed: a conserved population anchored at the group midpoint, $b(z) = 1 + (b_c - 1)\,D(z_c)/D(z)$ (Eq. (13)). Dash-dot, on BGS only: the $r < 19.5$ flux-limit bias, anchored at the same midpoint.

3.3 The prior's yardstick: the neighbor rms

The prior's quantity to compare with this tilt is the rms change between neighbors of Eq. (8). Table 2 evaluates $\sigma_b\sqrt{2\,(1 - e^{-\Delta z/L})}$ at several (σ_b, L) pairs and for the conserved population of Table 1.

At $\sigma_b = 1\%$ with $L = 0.1$ the prior allows neighbor differences of 0.6% over $\Delta z = 0.02$, matching the conserved drift. At $\sigma_b = 3\%$ with $L = 0.05$ the allowance is several times that drift, and at $\sigma_b = 10\%$ or 20% it reaches 8–27%, beyond any bias evolution identified in this note.

3.4 What breaks conservation in a selected sample

Four mechanisms break the conserved-population assumptions.

Threshold drift. The target selection holds a roughly constant comoving density until the flux limit bites; past that redshift the sample keeps only its brighter members, which sit in

Case	over $\Delta z = 0.02$	over $\Delta z = 0.05$
$\sigma_b = 3\%$, $L = 0.05$	2.4%	3.4%
$\sigma_b = 1\%$, $L = 0.1$	0.6%	0.9%
$\sigma_b = 5\%$, $L = 0.02$	5.6%	6.8%
$\sigma_b = 10\%$, $L = 0.05$	8.1%	11.2%
$\sigma_b = 20\%$, $L = 0.02$	22.5%	27.1%
conserved population	0.3–0.7%	0.8–1.7%

Table 2: The rms change of the bias deviation between two bins, $\sigma_b\sqrt{2\,(1 - e^{-\Delta z/L})}$, at several (σ_b, L) pairs, and for the drift of a conserved population (the per-group range of Table 1).

more massive halos and are more biased. The resulting bias change is smooth in redshift, with L comparable to the group width.

Color-selection features. The selection cuts in color space can respond quickly in redshift when a spectral feature crosses a filter edge. The sharpest common feature is the 4000 Å break: a filter edge 100–200 Å wide maps to $\Delta z \approx 0.03$–0.05 at that break, and photometric scatter smears the transition further. Its amplitude in bias is unknown.

Mergers. Two members that merge become one, so the number of galaxies is not conserved even before any selection acts. The size of this effect on these samples is unknown.

Emission-line selection. An ELG enters the sample while its emission lines are strong, so the membership follows the star-formation phase of each galaxy rather than one conserved set of objects.

How much each of these moves the bias, and over which Δz, is not known from theory alone; it is a question about the measured selection function of each sample.

3.5 What the measurements say

The mechanisms above can be weighed against published measurements. Each statement below carries a tag: *measured* means it is read from the cited source; *derived* means it is computed here from measured inputs; *assumed* means it is a judgment not yet backed by a measurement. Equation (13) itself is from Fry [1], which has no arXiv record; Tegmark and Peebles [2] (section "After the galaxy formation epoch") rederive it. Appendix G summarizes each paper below: the sample, the method and the numbers.

Massive red galaxies follow the conserved shape to the precision of the published tests. The passive-evolution analyses find merger corrections of a few percent per Gyr (measured [3, 4, 5]), and the DESI-like LRG bias of [6] stays within $\pm 2.3\%$ of the constant clustering amplitude $b = 1.5/D$ at a bin spacing of $\Delta z \approx 0.1$ below $z = 0.83$, with the highest bin $6.4 \pm 6.4\%$ high (measured biases; residuals derived here).

The selections drift. The LRG cuts hold the comoving density constant at $0.4 < z < 0.8$ and hand over to a fixed flux limit beyond (measured [7]); through the threshold-bias relation of [12] (measured), that flux limit makes the LRG3 bias drift by 6–11% per $\Delta z = 0.1$ on top of the B/D shape (derived), matching the low end of the measured highest-bin offset. The

magnitude-limited BGS (measured limit, [13]) drifts by 10–25% across its group through the same mechanism (derived).

The ELG bias shape is stable: HOD fits in two bins covering $0.8 < z < 1.6$ agree within one sigma (measured [15]). The one short-separation mechanism found in any tracer is the set of redshift windows where the [OII] 3727 Å doublet, which carries the ELG redshift, lands on sky emission lines and loses signal-to-noise, especially at $z > 1.5$ (mechanism measured [14]): each window is $\Delta z \approx 0.001$–0.01 wide, and its amplitude in bias is assumed, of order 1–2%: a judgment from the weak dependence of ELG bias on [OII] flux, not a measured bound.

None of the measurements collected here resolves bias below $\Delta z \approx 0.1$. The finest binning is the $\Delta z \approx 0.1$ of [6]; [15] use 0.3 and 0.5; and the lensing tomography of the same samples [9, 10, 11], built on the photometric samples of [8], uses about 0.15 with one free bias per bin. Below $\Delta z \approx 0.1$ the thin-bin statements of this note rest on the continuity equation and on the selection arguments.

3.6 Drift versus loss of memory between bins

Every deviation found above, except the sky-line windows, is a monotonic drift: the conserved-population tilt, the LRG3 magnitude-limit drift and the BGS flux-limit drift each move in one direction across their group. A monotonic drift is a coherent, long-correlation deformation, a tilt; it is not bin-to-bin decorrelation, so it does not argue for a short L. Within the exponential kernel this note analyzes, the measured drifts point to the long end of the scanned L grid, and only the sky-line windows point to the short end.

3.7 The scan and the plausible region

The forecast scans

$$L \in \{0.02,\ 0.05,\ 0.1,\ 0.2\} \quad \text{(in redshift)}, \qquad \sigma_b \in \{1\%,\ 3\%,\ 5\%,\ 7\%,\ 10\%\}, \tag{15}$$

with baseline $L = 0.05$ and $\sigma_b = 3\%$ (a provisional choice: the measurements of Section 3.5 can still move it), and the plausible region $L = 0.02$–0.1, $\sigma_b = 1$–5%. The upper end of L is physical, not numerical: a sample's flux limit and color cuts change which galaxies enter the sample as redshift grows, and that change of galaxy mix happens over $\Delta z \sim 0.1$, so bias deviations should not be assumed coherent over separations much beyond that: $L \lesssim 0.1$.

In Table 2, the baseline allows neighbor differences of 2.4% over $\Delta z = 0.02$, several times the conserved drift; the gentlest corner of the plausible region ($\sigma_b = 1\%$, $L = 0.1$) matches that drift, and the sharpest ($\sigma_b = 5\%$, $L = 0.02$) allows roughly eight times it. The rows $\sigma_b = 10\%$ and 20% show what the conserved-population comparison rules out: neighbor differences of 8–27%. Both sit above the plausible region, and $\sigma_b = 20\%$ is outside the scanned grid.

Shape freedom costs dark-energy information: the Fisher matrix spends part of the data constraining the bin biases instead of w_0 and w_a; Section 4 measures the cost across the scanned grid. Relative to a conserved population, the figure of merit computed anywhere in the plausible region except its gentlest corner is pessimistic.

This comparison is not a proof, because a DESI sample is not conserved: the selection, not gravity, decides which galaxies enter each bin.

4 Results

The forecast behind the scan is the FoM(w_0, w_a) of the DESI clustering band powers alone (C_ℓ^{gg}, no shear), computed non-Limber with the ultraviolet-fraction Fourier scale cut at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ of [16], which follows DeRose and Chen [17]. The cut keeps a band while the ultraviolet fraction $|C_\ell^{\mathrm{UV}}|/(|C_\ell^{\mathrm{UV}}| + |C_\ell^{\mathrm{IR}}|)$ stays below $\alpha = 0.05$, where the two pieces are the band power computed from the power spectrum above and below the split $k_{\max}/D(z)$. The figure of merit is the usual inverse area of the (w_0, w_a) error ellipse, $\mathrm{FoM} = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$: larger is better. Four thin-bin layouts are scanned: the $\Delta d_{\mathrm{com}} = 62$ Mpc layout (bins of equal comoving width) and the fixed-width layouts with $\Delta z = 0.02$, 0.05 and 0.2. In each layout the grid of Eq. (15) is bracketed by the tied and independent models.

Forecast setup. The survey covers 12,300 square degrees. The data vector is the project's measured DESI clustering block: the band powers of every bin auto-spectrum plus the measured adjacent cross-bin pairs (76 autos and 97 pairs on the $\Delta d_{\mathrm{com}} = 62$ Mpc layout), in 53 bands of width $\Delta\ell = 10$ over $\ell = 15$–544. The covariance is the source forecast's Gaussian matrix (cosmic variance plus shot noise), computed for 14,000 square degrees, with the inverse rescaled to the survey area. The model is linear bias with linear Kaiser redshift-space distortions; a per-group velocity dispersion smears each bin's $n(z)$ (the Fingers-of-God convolution); magnification is not modeled. The Fisher matrix varies seven cosmological parameters ($10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0, w_a, all with flat priors, at the project examples' fiducial values) plus the per-bin biases and the per-group velocity dispersions, the latter with their Gaussian priors of widths 30–145 km/s. Derivatives are finite differences; on $\Delta z = 0.2$ they are recomputed at four step sizes and two CAMB accuracy settings, and the FoM agrees to 0.6% across the steps, the numerical side of the Fisher approximation's validity.

Preliminary: linear bias only. The forecast models the galaxy field with linear bias alone, with no magnification; the absolute FoM values will move when higher-order bias terms enter the model. A second caveat is larger. The DESI covariance files are computed with each group's raw bias (1.34, 1.70, 0.84) while the data vector uses $b_i = B_g/D(\bar{z}_i)$, which understates the high-redshift variances and inflates every layout's absolute FoM by a factor of about 2.4–3.2; the project README quantifies the mismatch, and the ratios between the layouts are unchanged by it. Until the covariance is rebuilt with the matching bias, the absolute values here are upper bounds of a rigid linear-bias model, and the content of this section is the comparison between bias priors on a fixed layout, which holds the layout, the data vector and the covariance fixed and changes only the prior.

Figure 2 places every scanned (L, σ_b) between its layout's two caps. At the baseline ($\sigma_b = 3\%$, $L = 0.05$) the four layouts give FoM = 11.8, 8.05, 2.01 and 0.779, which is 0.36, 0.40, 0.24 and 0.41 of the respective tied values. At $\sigma_b = 1\%$ the tied fraction spans 0.45–0.79 across layouts and correlation lengths. At $\sigma_b \geq 5\%$ the $\Delta d_{\mathrm{com}} = 62$ Mpc and $\Delta z = 0.05$ layouts sit near their independent values: most of the information the tying provided is already spent. Two scanned points exceed the LSST Y1 3x2pt reference of 17.3 (the dashed line), both on the $\Delta d_{\mathrm{com}} = 62$ Mpc layout at $\sigma_b = 1\%$: FoM = 18.7 at $L = 0.1$ and 20.8 at $L = 0.2$. The reference is the `lsst_y1` project's 3x2pt Fisher forecast in the DESC Science Requirements Document Year 1 setup [18], with the figure of merit defined as here.

Figure 3 resolves the same numbers on the (L, σ_b) grid. L matters where neighboring bins

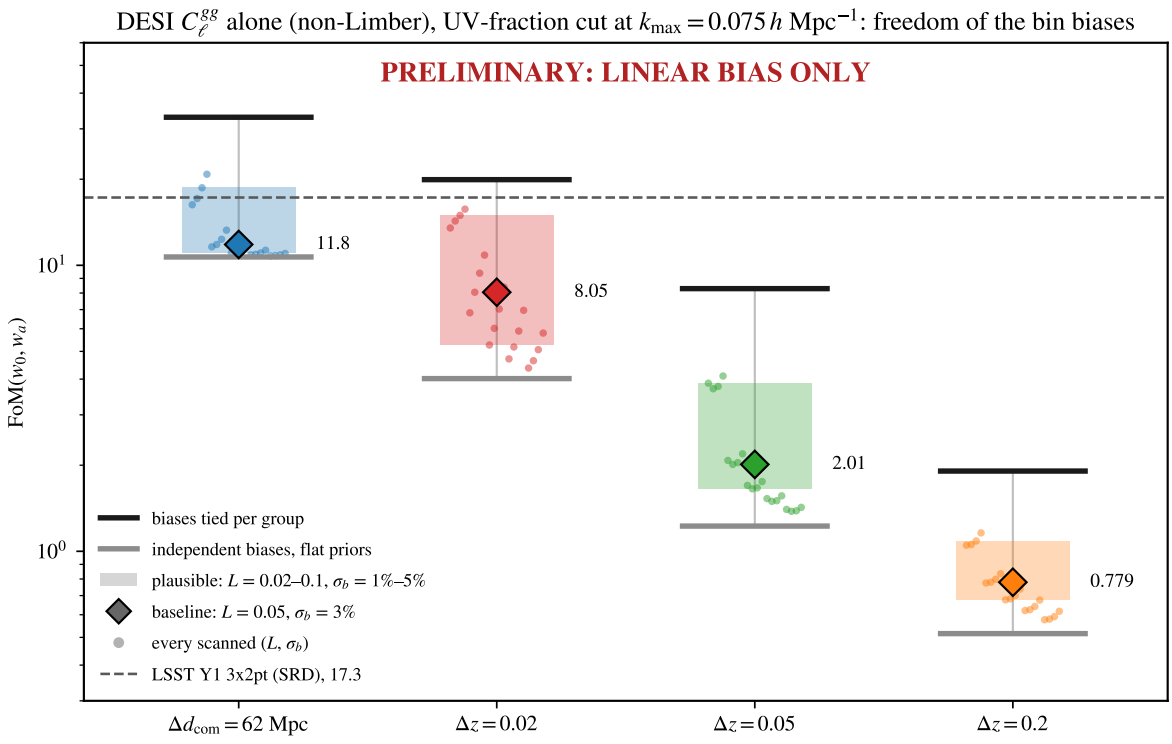


Figure 2: The FoM(w_0, w_a) of each thin-bin layout: the tied model (black cap), independent biases (gray cap), every scanned (L, σ_b) point (dots), the plausible region (σ_b = 1–5%, L = 0.02–0.1; band), the baseline (diamond), and the LSST Y1 3x2pt reference at 17.3 (dashed).

sit much closer than L, so that the correlation $e^{-\Delta z/L}$ stays high and lengthening L keeps converting bin-to-bin freedom back into effective tying. Both thin layouts are in that regime: the $\Delta d_{\rm com} = 62$ Mpc bins are $\Delta z = 0.016$–0.030 apart between $z = 0.2$ and 1.3, and on the $\Delta z = 0.02$ layout $e^{-\Delta z/L}$ runs from 0.37 at $L = 0.02$ to 0.90 at $L = 0.2$. At $\sigma_b = 1\%$ both layouts respond: the $\Delta z = 0.02$ FoM rises from 13.5 to 15.7 across the L grid, and the $\Delta d_{\rm com}$ FoM by 28%, from 16.3 to 20.8. At $\sigma_b \geq 3\%$ the $\Delta z = 0.02$ layout keeps a strong response (6.8 to 10.9 at 3%) while $\Delta d_{\rm com} = 62$ Mpc sits close to its independent floor of 10.69, which no prior choice moves. In the two wide layouts the bins are as far apart as L or farther, the neighbor correlation is low everywhere on the grid, and L moves the FoM only weakly.

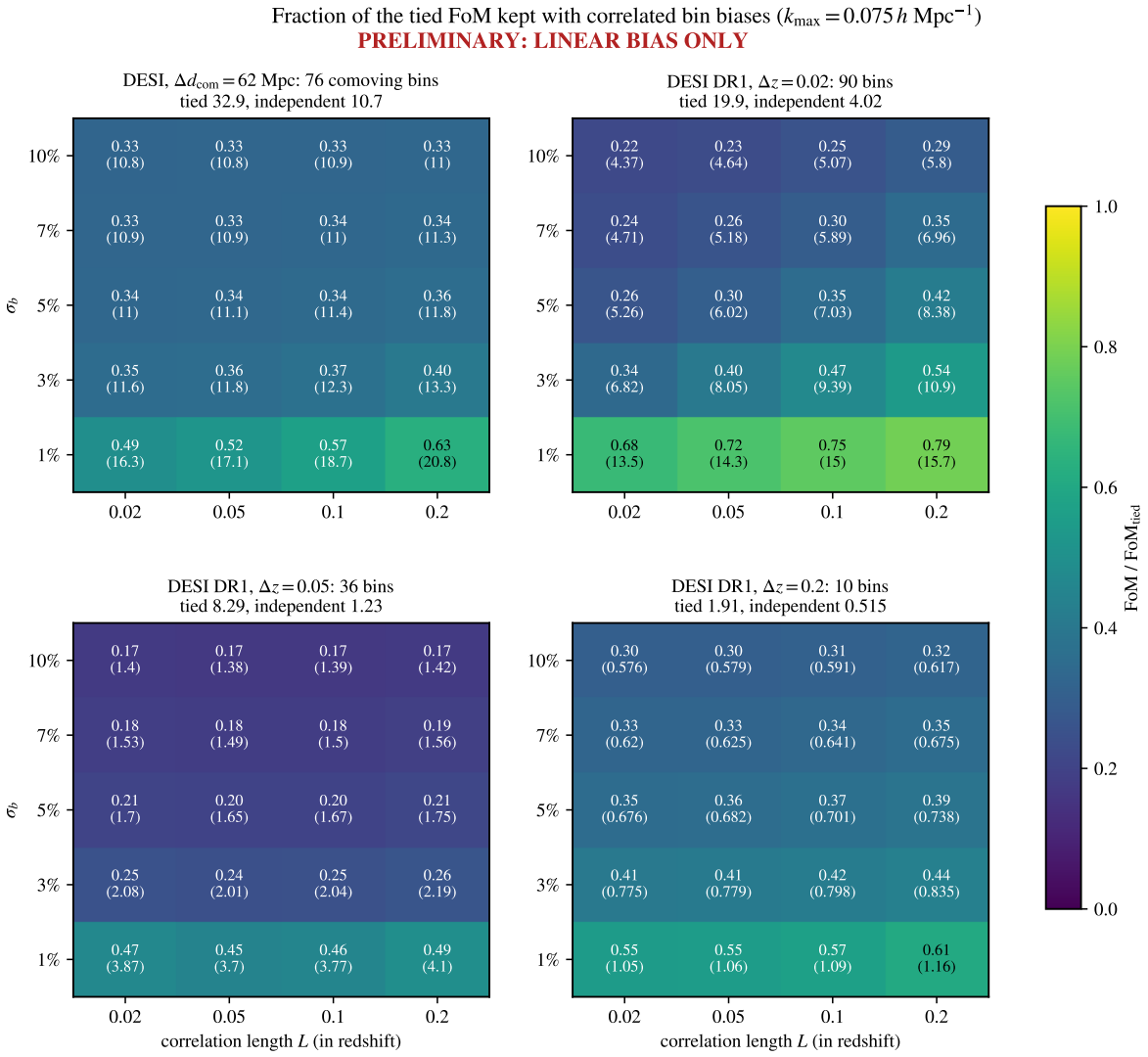


Figure 3: FoM/FoM$_{\rm tied}$ across the scanned (L, σ_b) grid, one panel per layout, with the absolute FoM in parentheses.

Figure 4 tests whether one number controls the cost. Its horizontal axis is $\sqrt{2}\,\kappa = \sigma_b\sqrt{2/L}$, the rms neighbor change per unit $\sqrt{\Delta z}$ of Eq. (10). The points do not collapse onto a single

curve per layout. The reason is that the roughness controls only the short-separation jumps: at fixed κ, a longer L also carries a larger wander $\sigma_b = \kappa\sqrt{L}$, so the bias can drift coherently over the whole group, a freedom the roughness alone does not capture. The Fisher matrix pays for both kinds of freedom, so no single-variable description of the grid should be expected.

The scan reaches its two limits numerically, as Appendix C.4 says it must. At $\sigma_b = 0.01\%$ with $L = 0.05$ the four layouts give FoM = 32.93, 19.93, 8.285 and 1.907, against tied values 32.94, 19.93, 8.288 and 1.908; at $\sigma_b = 1000\%$ they land on the independent values 10.69, 4.016, 1.225 and 0.5154 exactly.

5 The memory model

5.1 Why a second model

The kernel model carries one amplitude and one length, and Section 3.6 found that the measured deviations are not the kind it describes best. Every deviation found, except the narrow sky-line windows, is a line-like drift across its group: the BGS flux limit drifts by 10–25% across the group, the LRG3 magnitude limit by 6–11% per $\Delta z = 0.1$, and a conserved population by about 1% per $\Delta z = 0.05$ (Section 3.5).

The kernel model can admit such a drift, but only at a price. To pass a 20% drift with $L \leq 0.1$ it needs $\sigma_b = 10$–15%, and that spread also allows roughly 10% scatter between neighboring bins (Table 2), which no identified mechanism produces. One σ_b controls both the drift and the scatter, so the kernel cannot say "large smooth drift, small local scatter". The memory model separates the two. The left panel of Figure 5 shows what the kernel model assumes a galaxy population does; Figure 6 shows what the measured physics does.

5.2 Why the kernel forgets

The kernel's limitation is built into its correlation, $e^{-|z_i - z_j|/L}$, which decays with separation: at $L = 0.05$, bins 0.15 apart share $e^{-3} \approx 0.05$ of their deviation. Prediction makes the decay concrete. Appendix D shows that the best guess of the next bin given all earlier ones is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; r_k\, \delta_k : \tag{16}$$

only the nearest neighbor matters, and the influence of any one bin on the rest of the chain fades as the product of the neighbor correlations, roughly r^n after n steps. The kernel forgets.

5.3 The model, and why it remembers

The memory model adds one number per group: a tilt. The bias keeps the form of Eq. (2), and the deviation splits into a line and a residual,

$$b_i \;=\; \frac{B_g}{D(z_i)}\,\big(1 + \delta_i\big), \qquad \delta_i \;=\; s_g\, v_i + \varepsilon_i, \qquad v_i \;=\; z_i - \bar{z}_g, \tag{17}$$

where $\bar{z}_g$ is the mean of the group's bin redshifts, s_g is a free tilt per group with a flat prior (the same status as B_g), and the residual ε_i carries the exponential kernel of Eq. (3), $\langle \varepsilon_i \varepsilon_j \rangle = \sigma_b^2\, e^{-|z_i - z_j|/L}$.

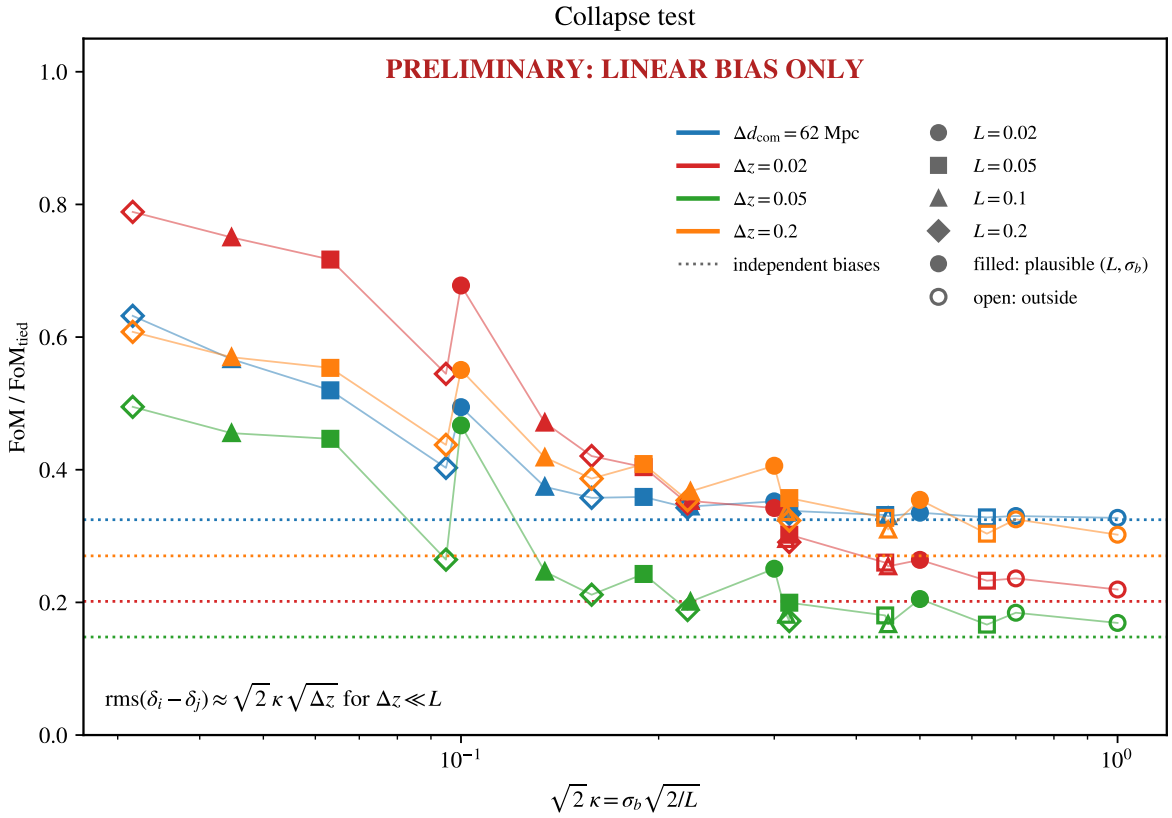


Figure 4: FoM/FoM$_{\rm tied}$ against the roughness combination $\sqrt{2}\,\kappa = \sigma_b\sqrt{2/L}$, which sets the rms neighbor change at small separations, $\sigma_b\sqrt{2\,\Delta z/L}$ (Eq. (10)): for $\sqrt{2}\,\kappa = 0.1$, the bias changes by 1% between bins $\Delta z = 0.01$ apart, and 1.4% for $\Delta z = 0.02$. Colors distinguish the four layouts, with each layout's independent fraction FoM$_{\rm indep}$/FoM$_{\rm tied}$ dotted; marker shapes give L (circle 0.02, square 0.05, triangle 0.1, diamond 0.2). Markers are filled when (L, σ_b) lies inside the plausible region, $\sigma_b = 1$–5% and $L = 0.02$–0.1, and open outside it. Plausible means inside the following edges, not a probability. The lower edge $\sigma_b = 1\%$ is derived: a conserved population drifts from B/D by about 1% per $\Delta z = 0.05$ (Section 3.2), so a narrower prior would be tighter than population evolution alone. The upper edge $\sigma_b = 5\%$ rests on measurements: the measured LRG residuals around B/D are at most 2.3%, with errors of 1–3.4%, at $\Delta z \approx 0.1$ spacing ([6]; Section 3.5); the collected measurements of Appendix G place the upper envelope of the smooth LRG and ELG deviations at about 5%; and the neighbor-rms comparison of Section 3.3 shows that $\sigma_b \geq 10\%$ lets neighboring bins differ by 8–27%, which no identified mechanism produces, so 10% and 20% are excluded from the region and 10% stays in the scan only as the edge of the grid. The upper edge $L = 0.1$ is a modeling choice: the argument of Section 3.7 that a sample's flux limit and color cuts change the galaxy mix over $\Delta z \approx 0.1$, although the measured drifts are smooth and favor the long end of the L grid (Section 3.6), the case the memory model of Section 5 addresses. The lower edge $L = 0.02$ is the thinnest bin spacing of the scanned layouts: below it, neighboring bins would decorrelate faster than the bins are spaced.

To see what the tilt changes, give the slope a Gaussian prior of variance τ^2 first (the flat prior is the limit $\tau \to \infty$). The deviations then have covariance

$$\mathrm{Cov}(\delta_i, \delta_j) \;=\; \tau^2\, v_i\, v_j \;+\; \sigma_b^2\, e^{-|z_i - z_j|/L}. \tag{18}$$

The new term does not decay with separation. Bins on the same side of the group center ($v_i v_j > 0$) are positively correlated, bins on opposite sides negatively. For separations much larger than L the kernel term is gone and the correlation coefficient tends to

$$\rho_{ij} \;\longrightarrow\; \frac{\tau^2 v_i v_j}{\sqrt{\big(\tau^2 v_i^2 + \sigma_b^2\big)\big(\tau^2 v_j^2 + \sigma_b^2\big)}} \;\longrightarrow\; \pm 1 \quad \text{when } \tau|v| \gg \sigma_b : \tag{19}$$

the two ends of a group remember each other through the shared slope (Appendix E derives both steps).

Prediction shows the same memory. The best guess of the next bin becomes

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_1, \ldots, \delta_k\big] \;=\; \hat{a} + \hat{s}_g\, v_{k+1} \;+\; r_k\, \big(\delta_k - \hat{a} - \hat{s}_g\, v_k\big), \tag{20}$$

where $(\hat{a}, \hat{s}_g)$ is the best-fitting straight line through all the past bins, weighted by C^{-1} (it needs at least two past bins). Every past bin now enters, through the fitted line; only the wiggle around the line stays Markov. Statistics calls Eq. (20) the universal-kriging predictor, prediction with an unknown linear mean; Appendix E proves it.

Figure 5 makes the difference visible on the BGS bins at $\Delta z = 0.02$. In the left panel, kernel draws at $\sigma_b = 10\%$, about what the kernel needs to admit a drift of BGS size, zigzag and forget: bins 0.1 apart are unrelated, and no draw looks like the thick flux-limit drift. In the right panel, memory draws with a tilt plus $\sigma_b = 2\%$ are lines with small wiggles, and the drift looks like a typical draw.

5.4 Why the memory model differs physically

A selected sample changes along a group in two physically different ways. The first kind is coherent: an evolution of the whole sample that moves every bin the same way. A flux or magnitude threshold that rises with distance, the debiasing of a conserved population, and the drift of the halo-mass threshold at fixed number density all act like this. The second kind is local: features of the selection that change bins independently and are forgotten over a short range, such as the sky-line windows and filter-edge crossings of Section 3.4. The exponential kernel knows only the second kind, so it must inflate σ_b to admit the first. The memory model gives the first kind to the tilt and the second to the kernel.

One straight line is enough for the coherent kind because the drifts are smooth functions of redshift, set by $D(z)$ and the distance modulus, which change on Δz of order one. Over a group span of 0.2–0.5 the first-order Taylor term dominates. Table 3 measures this (derived): the best-fitting straight line through each group's conserved-population deviation, evaluated at the $\Delta z = 0.05$ bin centers with equal weights, and the residual the line leaves. A line removes the conserved-population drift to better than 0.3%.

The flux-limit drift is larger and not exactly straight. For BGS ($r < 19.5$, with r the apparent-magnitude band rather than a correlation coefficient; no K-correction; the threshold-bias fit of [12] with $M_* = -20.44$ used without evolution; magnitudes in the $M - 5\log_{10} h$

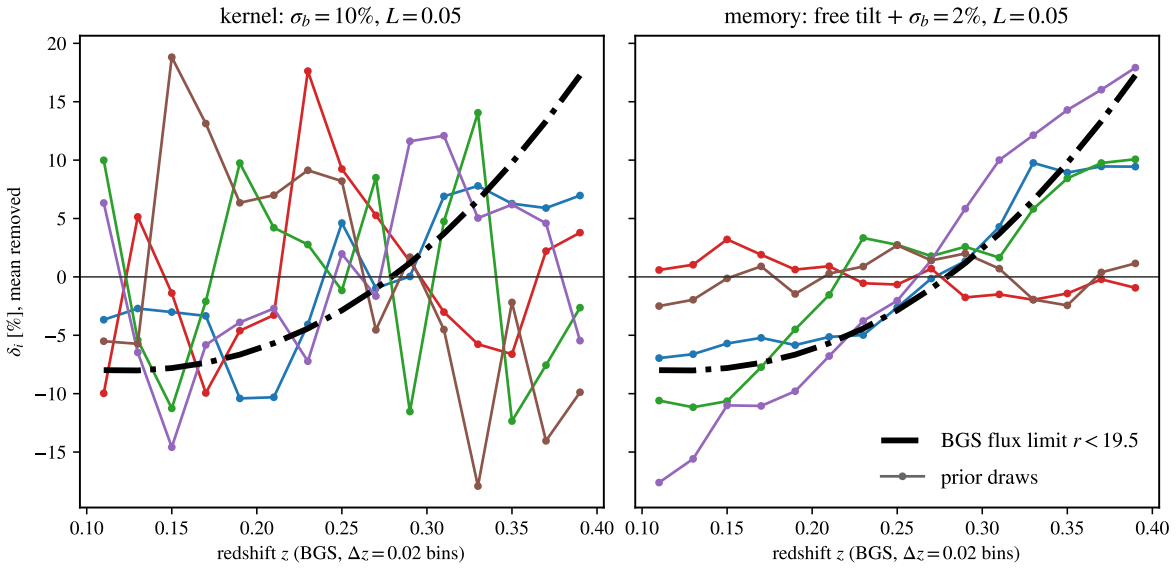


Figure 5: Random prior draws of the deviations of the 15 BGS bins at $\Delta z = 0.02$, each draw's mean removed (B_g is free). The thick dash-dot line is the BGS flux-limit drift. Left: the kernel model with $\sigma_b = 10\%$ and $L = 0.05$. Right: the memory model, a tilt plus $\sigma_b = 2\%$ and $L = 0.05$. The slope is a free parameter of the model; for display only, the drawn slopes come from a Gaussian of width 0.6 per unit redshift.

convention that matches M_*; and the group amplitude set so the mean deviation vanishes), the limiting magnitude runs from -18.56 to -21.24 between the first and last bin centers, the bias from 1.09 to 1.54, and the deviation spans 22.3% with a fitted slope of 89% per unit redshift. The line leaves 2.4% rms and 3.3% at most: the drift is convex, because the bias grows exponentially with the luminosity threshold (through the magnitude). So for BGS the memory model's residual scatter must be about 2–3%, or the model needs a quadratic term. The estimate is crude: no K-correction, and no evolution of the threshold-bias fit.

Figure 6 carries both statements. In the top panel every conserved deviation is a straight line, and the thin gray fitted lines sit on top of them. In the bottom panel the conserved residuals are flat at zero, while the BGS flux limit leaves a $\pm 3\%$ U shape: that U is the curvature the memory model's σ_b must absorb.

The two models use the same symbol σ_b for different physical quantities. In the kernel model, σ_b is the total departure of a bin from the B_g/D trend: drift plus scatter. In the memory model, σ_b is the departure from the group's own straight line: curvature plus local scatter. The physically plausible values differ accordingly: about 0.1% for a conserved population (Table 3), 1–2% for the local selection features (Section 3.5), and 2–3% for the BGS curvature.

The honest limit is unchanged: none of the measurements collected in Section 3.5 resolves bias below $\Delta z \approx 0.1$. The memory model rests on the same evidence as the kernel model; it encodes that evidence in a form closer to the physics.

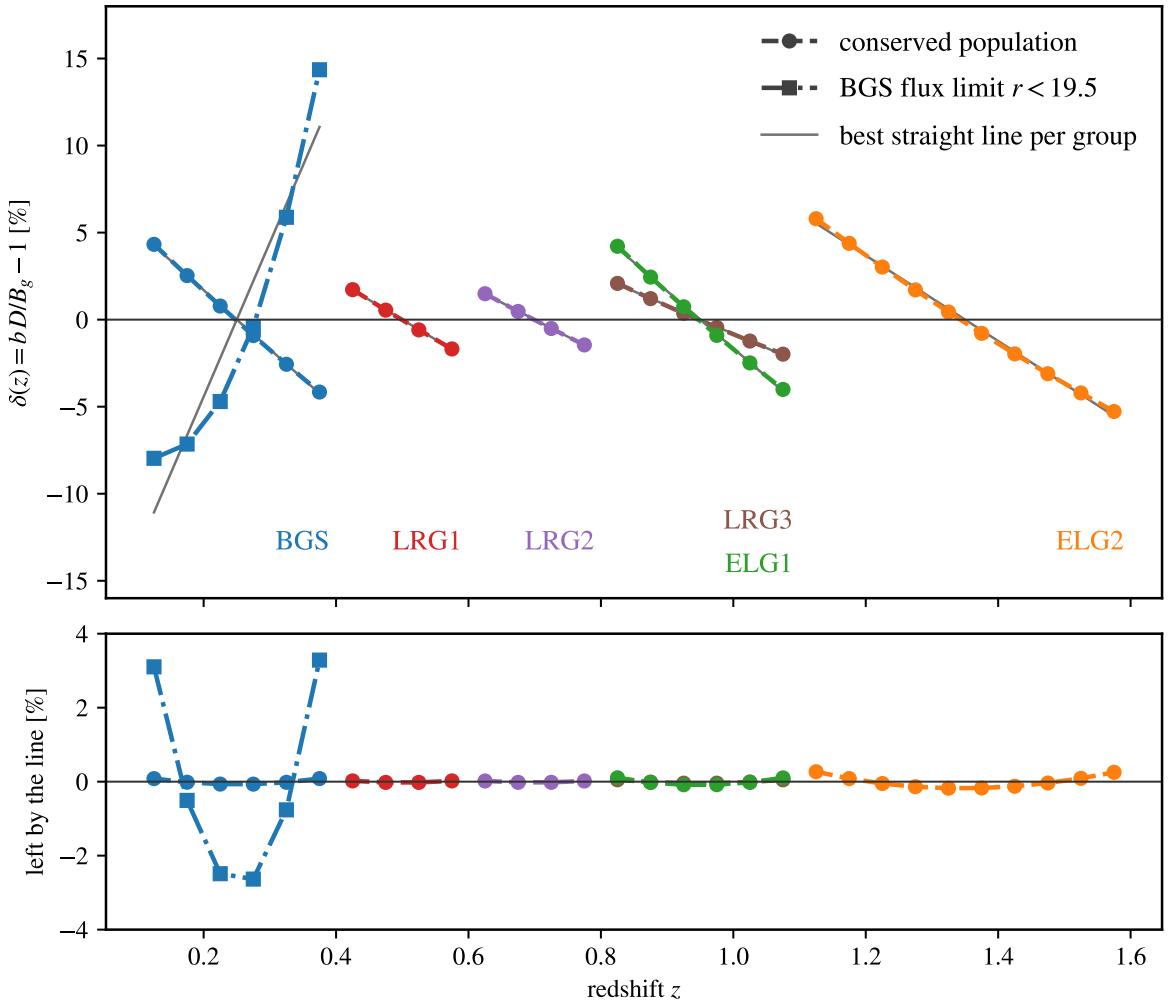


Figure 6: Top: the deviation $\delta(z) = b\,D/B_g - 1$ at the $\Delta z = 0.05$ bin centers, with B_g set so the mean deviation vanishes. Dashed with circles: the conserved population, every group. Dash-dot with squares: the BGS flux limit. Thin gray: the best straight line per curve. Bottom: what the line leaves. The conserved residuals are invisible (under 0.3%); the BGS flux-limit residual is convex, about ±3%. This is the figure behind Table 3.

Group	z range	span of δ	slope per unit z	residual rms	residual max
BGS	0.1–0.4	8.49%	−34.0%	0.062%	0.084%
LRG1	0.4–0.6	3.41%	−22.7%	0.020%	0.020%
LRG2	0.6–0.8	2.94%	−19.6%	0.018%	0.019%
LRG3	0.8–1.1	4.07%	−16.3%	0.038%	0.052%
ELG1	0.8–1.1	8.23%	−32.9%	0.077%	0.104%
ELG2	1.1–1.6	11.09%	−24.6%	0.158%	0.270%

Table 3: The best-fitting straight line through each group's conserved-population deviation $\delta(z)$ of Eq. (14), at the $\Delta z = 0.05$ bin centers with equal weights (derived; flat ΛCDM, $\Omega_m = 0.3$, amplitudes $B = 1.34$ for BGS, 1.70 for the LRGs and 0.84 for the ELGs, the reference at the group midpoint). The span is δ at the first center minus δ at the last; the residual columns describe what the line leaves.

5.5 What the prior charges

With the flat priors restored (slope variance $\tau^2 \to \infty$, and amplitude variance $W^2 \to \infty$ for B_g), the prior precision of a group's bin amplitudes becomes

$$\mathsf{P}_{\rm mem} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}, \qquad \mathsf{Q} = \mathsf{C}^{-1}, \qquad \mathsf{U} = [\,\mathbf{1},\; v\,], \tag{21}$$

the same construction as Eq. (12) with one more column: U is the $n \times 2$ matrix whose columns are the ones vector and the centered redshifts v. The penalty it charges is

$$\delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta \;=\; \min_{a,\,s_g} \big(\delta - a\,\mathbf{1} - s_g\,v\big)^{\mathsf{T}}\mathsf{Q}\big(\delta - a\,\mathbf{1} - s_g\,v\big)\,: \tag{22}$$

the prior subtracts the best-fitting straight line, weighted by C^{-1}, and charges only the residual. A monotonic drift of any size costs nothing; bumps around the line still cost, and still forget over the length L.

The structure of $\mathsf{P}_{\rm mem}$ separates local from global. Q is tridiagonal (Appendix C): a local object that links neighbors. The subtracted correction has rank two and is dense: it couples every bin to every other, because the fitted line listens to all of them. The kernel model already had one dense rank-one piece, the subtracted mean of Eq. (12); the memory model adds the slope. $\mathsf{P}_{\rm mem}$ annihilates both columns, $\mathsf{P}_{\rm mem}\mathsf{U} = 0$, is positive semidefinite, and has rank $n-2$ (Appendix E).

A group of one or two bins has $\mathsf{P}_{\rm mem} = 0$: an amplitude and a slope fix every bin, so the group is fully free. Among the scanned layouts this mostly affects $\Delta z = 0.2$, whose groups hold the fewest bins.

5.6 What the slope costs

The tilt has a physical price. In linear theory the clustering band powers of bin i scale as $b_i^2\,D(z_i)^2$, so the data measure the product $b\,D$, not the factors. Why a free tilt costs growth information takes one line: with $f = d\ln D/d\ln a$ the linear growth rate, $d\ln D/dz = -f(z)/(1+z)$, so inside a group

$$\frac{d\ln(bD)}{dz} \;=\; \frac{d\ln b}{dz} \;-\; \frac{f(z)}{1+z}. \tag{23}$$

A free tilt in b removes the channel that measures f through the amplitude evolution inside a group. The free amplitude B_g already handed each group's overall growth to the bias; the free tilt also hands over the relative growth inside a group.

Not everything is lost to the degeneracy. The redshift-space terms scale as $b\,f$, not as b^2, and measure f directly; the geometric distances and the shape of the power spectrum are untouched; and the comparison of growth between groups survives, because each group gives away only its own interior tilt, a statement this note does not test. This is why "tied plus free tilt" sits below the tied model, and why it should sit well above the independent one. The limit $\sigma_b \to 0$ of the memory model isolates the question: it measures how much of the figure of merit came from growth measured inside groups.

5.7 Results of the memory model

The memory model was scanned on the same grid as the kernel model: the same four layouts and the same FoM(w_0, w_a) of the DESI clustering band powers alone at $k_{\rm max} = 0.075\,h/$Mpc, with σ_b and L now describing the residuals around each group's free line. The caveat of Section 4 applies unchanged: these results are preliminary, linear bias only.

Layout	tied	tied + free tilt	memory, 1%	kernel, 1%	independent
$\Delta d_{\rm com} = 62$ Mpc	32.94	24.35 (0.74)	14.99	17.12	10.69
$\Delta z = 0.02$	19.93	13.34 (0.67)	11.39	14.29	4.016
$\Delta z = 0.05$	8.288	2.654 (0.32)	1.683	3.701	1.225
$\Delta z = 0.2$	1.908	0.5154 (0.27)	0.5154	1.056	0.5154

Table 4: FoM(w_0, w_a) at $L = 0.05$. The tied-plus-free-tilt column is the $\sigma_b \to 0$ limit of the memory model, with its fraction of the tied value in parentheses; the memory and kernel columns are both at $\sigma_b = 1\%$.

The price of the tilt alone is the distance between the first two columns of Table 4. One free tilt per group costs 26% of the tied FoM on $\Delta d_{\rm com} = 62$ Mpc, 33% on $\Delta z = 0.02$, 68% on $\Delta z = 0.05$, and 73% on $\Delta z = 0.2$, where the free amplitude and tilt already free every one- and two-bin group completely. That is the information the tied model drew from the evolution of $b\,D$ inside each group, the growth channel of Eq. (23). $\Delta z = 0.05$ leans on that channel far more than the thinner layouts; a plausible reading, offered as interpretation rather than an established result, is that the thinner layouts measure more through the redshift-space terms and the cross-bin structure, which a tilt does not absorb.

At the same σ_b the memory model sits below the kernel model, as it must: it has strictly more freedom. In Figure 7 the dashed memory curves run below the solid kernel curves at small σ_b. That is not the comparison that matters.

The comparison that matters pairs each model with the σ_b it needs to describe the physics of Sections 3 and 5.4: about 10% for the kernel model to admit the measured drifts, and about 1% (a conserved, LRG-like population) to 3% (the BGS curvature) for the memory model. At $L = 0.05$, $\Delta d_{\rm com} = 62$ Mpc gives memory 14.99 at 1% and 11.45 at 3% against kernel 10.81 at 10%; $\Delta z = 0.02$ gives 11.39 and 7.41 against 4.64; and $\Delta z = 0.05$ gives 1.68 and 1.33 against 1.38. Once the measured drifts are admitted, the memory model keeps more of the FoM on the two thinnest layouts, and about the same on $\Delta z = 0.05$.

Figure 8 resolves the memory grid. As in the kernel model, L matters where neighbors

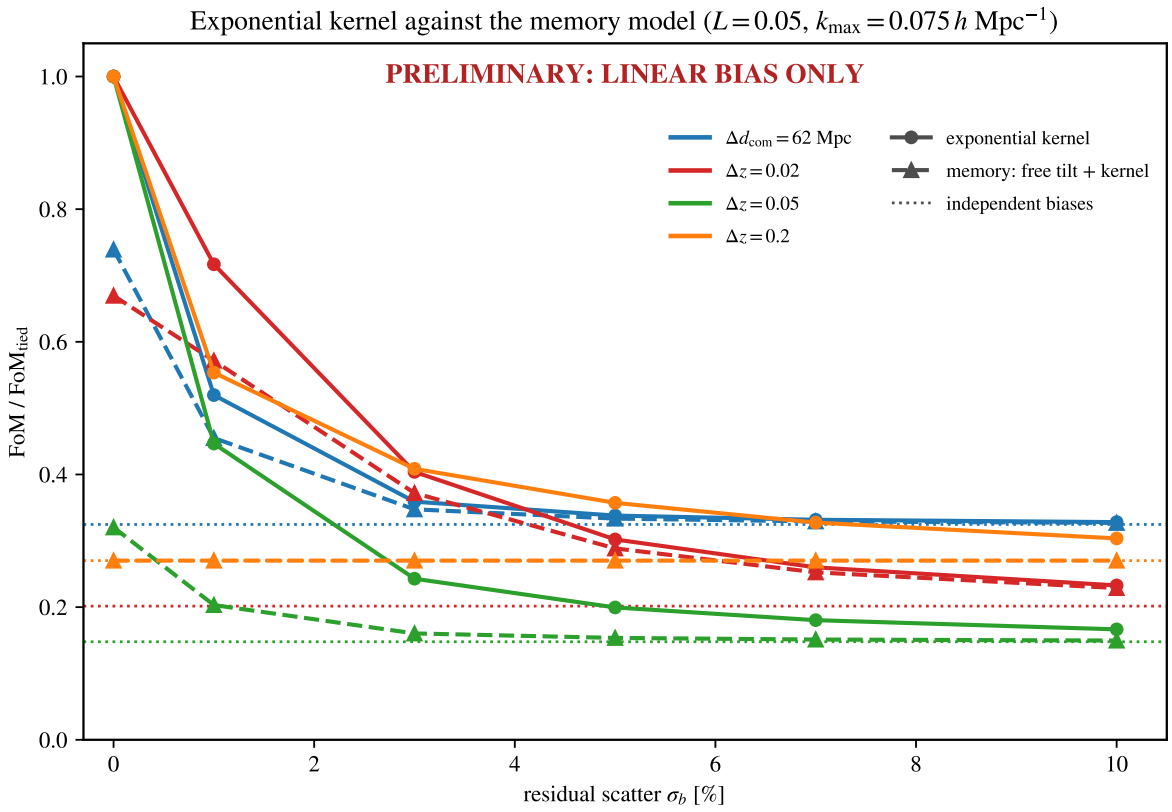


Figure 7: The two models against σ_b at $L = 0.05$, as the fraction of each layout's tied FoM. Solid lines with circles: the kernel model. Dashed lines with triangles: the memory model. Dotted: each layout's independent value. At $\sigma_b = 0$ the kernel model sits at one (the tied model) and the memory model at its tied-plus-free-tilt value. The dashed orange line of $\Delta z = 0.2$ lies on its dotted independent line.

sit much closer than L: at $\sigma_b = 3\%$ the $\Delta z = 0.02$ FoM runs from 6.1 at $L = 0.02$ to 10.0 at $L = 0.2$, and the $\Delta d_{\rm com} = 62$ Mpc layout responds at $\sigma_b = 1\%$ (14.0 to 18.2) before its independent floor takes over. $\Delta z = 0.2$ is degenerate by construction: every cell equals the independent value 0.5154, because each of its groups holds one or two bins, which the free amplitude and tilt free completely (Section 5.5).

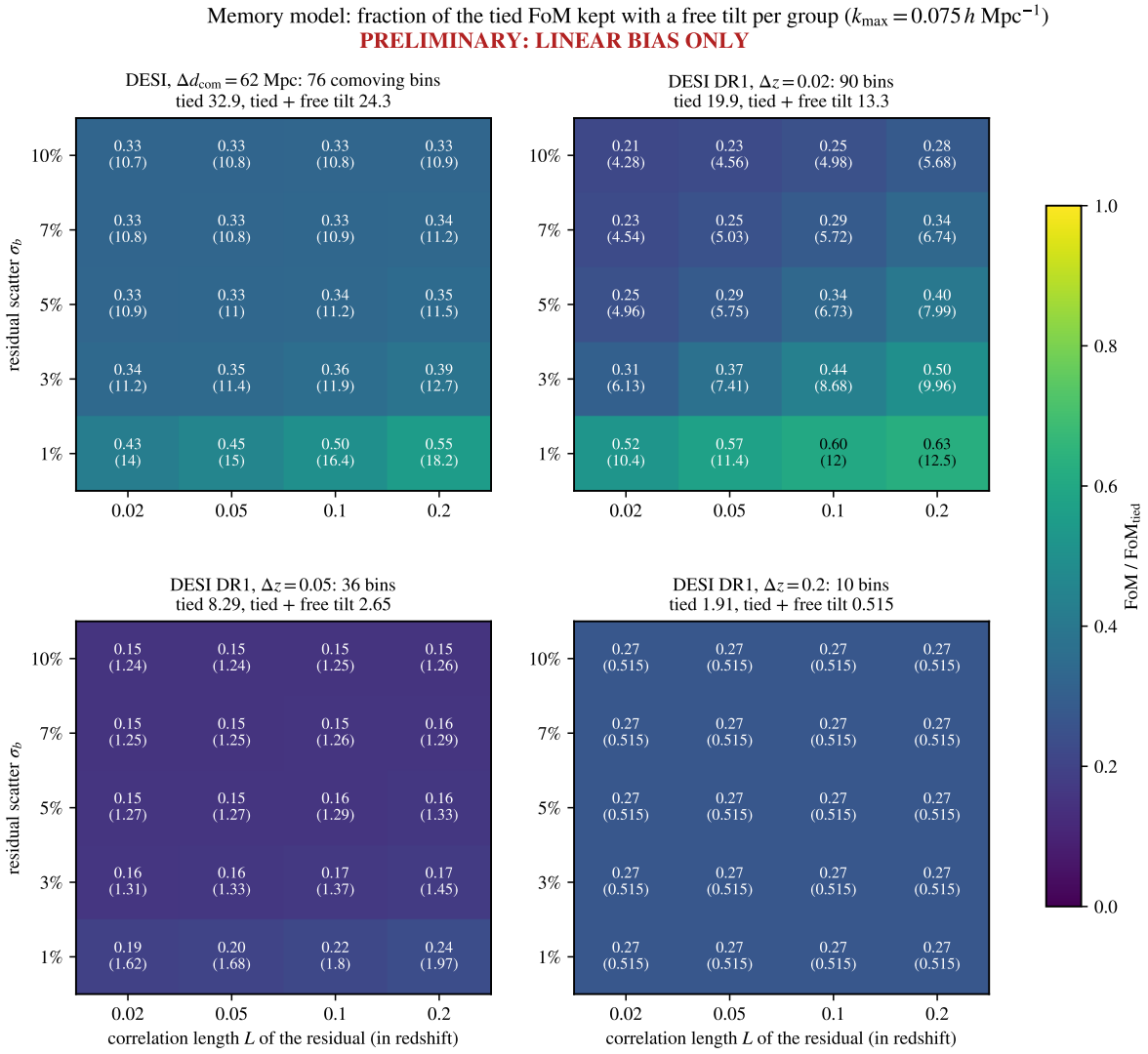


Figure 8: The memory model across the scanned (L, σ_b) grid, in the format of Figure 3: one panel per layout, with the absolute FoM in parentheses and each panel's tied-plus-free-tilt value in its title.

The scan reaches its limits numerically. At $\sigma_b = 1000\%$ the memory model lands on the independent values exactly (10.69, 4.016, 1.225, 0.5154), and at $\sigma_b \to 0$ on the tied-plus-free-tilt values of Table 4, as Section 5.8 says it must.

5.8 The limits

The memory model contains the same two endpoints as the kernel model, shifted by the tilt. As $\sigma_b \to 0$ the residuals vanish and every group is a rigid line with a free level and a free slope: the tied model plus a free tilt per group. As $\sigma_b \to \infty$ the prior charges nothing and the biases are independent. Between the endpoints, σ_b and L keep the same meanings as before, now applied to the wiggles around each group's line.

Summary

The intermediate bias model frees one amplitude B_g per tracer group and lets each bin deviate fractionally from the group's constant-amplitude trend $B_g/D(z)$, with deviations correlated over a redshift length L and a per-bin wander σ_b. In the Fisher matrix this becomes one projected precision P per group, added to the bin-amplitude block: P charges for differences between neighboring amplitudes but never for a uniform shift of the group. The unprojected inverse covariance links only neighboring bins directly (it is tridiagonal, even though the correlation itself reaches far); the projection adds one dense rank-one piece, the subtracted mean. The prior reduces to the tied model as $L \to \infty$ and to independent flat-prior biases as $\sigma_b \to \infty$. A second model adds a free tilt per group and reserves σ_b for the residuals around each group's line: the tilt alone costs 26–73% of the tied FoM, and at the σ_b each model needs to admit the measured drifts (about 1–3% with a tilt, about 10% without), the memory model keeps more of the figure of merit on the thinnest layouts.

A The projected precision for two bins

Everything essential already happens with two bins. Call their amplitudes β_1 and β_2, with common fiducial value B, prior spread s per bin, and correlation coefficient r between the two deviations. The prior covariance and its inverse are

$$\mathsf{C} \;=\; s^2 \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r \\ -r & 1 \end{pmatrix}. \tag{24}$$

(The inverse is checked by multiplying: the off-diagonal of $\mathsf{C}\,\mathsf{C}^{-1}$ is $r - r = 0$ and the diagonal is $(1-r^2)/(1-r^2) = 1$.)

Now examine the prior penalty $\chi^2_{\rm prior} = \Delta\beta^{\mathsf{T}}\,\mathsf{C}^{-1}\Delta\beta$, where $\Delta\beta = (\beta_1 - B,\ \beta_2 - B)^{\mathsf{T}}$ is the displacement from the fiducial. Two kinds of move matter.

A uniform shift. Take $\Delta\beta = a\,(1,1)^{\mathsf{T}}$: both bins move up by the same amount a. Working out the quadratic form with Eq. (24),

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{a^2}{s^2(1-r^2)}\,\big[\,(1-r)+(1-r)\,\big] \;=\; \frac{2a^2}{s^2\,(1+r)} \;\neq\; 0. \tag{25}$$

The plain prior charges for a uniform shift. But a uniform shift of all β_i is a change of B_g, a direction the tied model leaves completely free. If the intermediate model kept this penalty, it would constrain B_g and the comparison between the models would be unfair.

A shape change. Take $\Delta\beta = a\,(1,-1)^{\mathsf{T}}$: the bins move apart. The same algebra gives

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; \frac{2a^2}{s^2\,(1-r)}. \tag{26}$$

This is the penalty the prior is *meant* to apply: it grows without bound as $r \to 1$, where the two bins are forced to track each other, and it relaxes to the independent value $2a^2/s^2$ as $r \to 0$.

A number makes the contrast concrete. At $r = 0.670$ (the worked value of Section 2.2), the shape change costs $2a^2/(0.330\,s^2) = 6.1\,a^2/s^2$ while the uniform shift costs $2a^2/(1.670\,s^2) = 1.2\,a^2/s^2$. The correlated prior already charges five times more for the shape change than for the uniform shift; the projection below takes the uniform-shift charge all the way to zero.

The projected precision. The repair for the uniform shift is the projection P of Eq. (12). Appendix B explains where the formula comes from; here, compute it for two bins. Because the two rows of C^{-1} have the same sum, $\mathsf{C}^{-1}\mathbf{1} = \big[s^2(1+r)\big]^{-1}(1,1)^{\mathsf{T}}$ and $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} = 2/\big[s^2(1+r)\big]$. Substituting into Eq. (12) and simplifying each entry,

$$\mathsf{P} \;=\; \frac{1}{2s^2(1-r)}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; \frac{(\beta_1-\beta_2)^2}{2s^2(1-r)}. \tag{27}$$

The projected prior charges only for the *difference* of the two amplitudes. A uniform shift, $\beta_1 = \beta_2$, is now free, as in the tied model; a shape change keeps exactly the penalty of Eq. (26).

The denominator is not an arbitrary constant. With $\mathrm{Var}(\beta_i) = s^2$ and $\mathrm{Cov}(\beta_1,\beta_2) = s^2 r$,

$$\mathrm{Var}(\beta_1-\beta_2) \;=\; s^2 + s^2 - 2s^2 r \;=\; 2s^2(1-r)\,: \tag{28}$$

the projected penalty is the squared jump between the bins divided by the prior variance of that jump. The whole construction reduces to one sentence: *the prior constrains the jump, not the level.*

B Why the projection "sees" the uniform shift: three views

Equation (12) subtracts from C^{-1} a matrix built out of the single vector $v = \mathsf{C}^{-1}\mathbf{1}$: the subtracted piece is $v\,v^{\mathsf{T}}$ divided by the number $\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$. A matrix built from one vector acts along one direction only, so the subtraction removes the prior's information about exactly one direction, the uniform shift, and leaves everything else alone. This section derives the formula three times, each derivation adding its own intuition. Throughout, a group has n bins; $\Delta\beta$ and $\mathbf{1}$ are vectors with n entries.

B.1 View (a): subtracting the best-fitting uniform shift

Ask the following question: given a displacement $\Delta\beta$, how much of it can be explained as a pure change of the group amplitude? A change of B_g by c moves every β_i by c, so the candidate explanation is $c\,\mathbf{1}$, and the unexplained part should be judged by the prior metric. Minimize over the scalar c the quadratic form

$$\chi^2(c) \;=\; \big(\Delta\beta - c\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big). \tag{29}$$

Differentiating with respect to c (the two cross terms are equal because C^{-1} is symmetric),

$$\frac{d\chi^2}{dc} \;=\; -2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\big(\Delta\beta - c\,\mathbf{1}\big) \;=\; 0, \qquad \frac{d^2\chi^2}{dc^2} \;=\; 2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;>\; 0, \tag{30}$$

where the second derivative is positive because C^{-1} is positive definite, so the solution of the first equation is the minimum of $\chi^2(c)$. That solution is

$$a \;=\; \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}. \tag{31}$$

For a diagonal C with variances s_i^2 this reduces to $a = \sum_i \Delta\beta_i/s_i^2 \big/ \sum_i 1/s_i^2$, the familiar inverse-variance weighted mean; Eq. (31) is its generalization to correlated errors: generalized least squares for a constant.

Keep the dimensions straight: $\Delta\beta$, $\mathbf{1}$ and the residual $w = \Delta\beta - a\,\mathbf{1}$ are vectors with n entries; a is a single number. The residuals w satisfy one linear constraint (Eq. (33) below), so they span $n-1$ dimensions: one direction of the n-dimensional displacement space has been handed to the amplitude, and $n-1$ directions of genuine shape freedom remain.

B.2 View (b): a Pythagorean split of the penalty

Write the displacement as the best uniform shift plus a residual,

$$\Delta\beta \;=\; a\,\mathbf{1} + w. \tag{32}$$

The first-order condition of Eq. (30) says precisely that

$$\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; 0\,: \tag{33}$$

the residual is orthogonal to the uniform direction *in the* C^{-1} *metric*. (With correlated errors, "orthogonal" must be measured with the same matrix that measures lengths.) Substituting the split into the full penalty, the cross term is $2a\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}w = 0$, and

$$\Delta\beta^{\mathsf{T}}\mathsf{C}^{-1}\Delta\beta \;=\; a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \;+\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{34}$$

This is the Pythagorean theorem in the prior metric: the squared length of the displacement is the squared length of its uniform piece plus the squared length of its shape piece.

Now compare with the projection. Inserting Eq. (31) into the subtracted term of Eq. (12) shows that the subtraction removes exactly $a^2\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$, the uniform piece. What remains is

$$\Delta\beta^{\mathsf{T}}\mathsf{P}\,\Delta\beta \;=\; w^{\mathsf{T}}\mathsf{C}^{-1}w. \tag{35}$$

Two consequences follow immediately. First, a pure uniform shift has $w = 0$, so its penalty vanishes. Equivalently $\mathsf{P}\,\mathbf{1} = 0$: the projected matrix sends the uniform direction to zero and charges nothing for it. Second, a pure shape change (a displacement that already satisfies Eq. (33)) has $a = 0$ and keeps its original penalty untouched. The projection removes one direction and alters nothing else.

B.3 View (c): marginalizing a flat-prior amplitude

The third view explains why the projection, rather than some other uniform-shift repair, is the *correct* prior. Model the group amplitudes as

$$\beta \;=\; B\,\mathbf{1} + \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, \mathsf{C}), \tag{36}$$

with a flat prior on the scalar B. This is literally the intermediate model of Eq. (2), linearized around the fiducial. The prior density of β is obtained by integrating B out:

$$p(\beta) \;\propto\; \int dB\, \exp\Big[-\tfrac{1}{2}\big(\beta - B\,\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\,\mathbf{1}\big)\Big]. \tag{37}$$

The exponent is quadratic in B. Writing $S = \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}$ and completing the square,

$$\big(\beta - B\mathbf{1}\big)^{\mathsf{T}}\mathsf{C}^{-1}\big(\beta - B\mathbf{1}\big) \;=\; S\,\big(B - B_*\big)^2 \;+\; \beta^{\mathsf{T}}\mathsf{C}^{-1}\beta \;-\; \frac{\big(\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta\big)^2}{S}, \qquad B_* = \frac{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\beta}{S}. \tag{38}$$

The B integral is a Gaussian centered on B_* with width $1/\sqrt{S}$; it contributes only a β-independent constant $\sqrt{2\pi/S}$. What survives in the exponent is

$$p(\beta) \;\propto\; \exp\Big[-\tfrac{1}{2}\,\beta^{\mathsf{T}}\mathsf{P}\,\beta\Big], \qquad \mathsf{P} \;=\; \mathsf{C}^{-1} - \frac{\mathsf{C}^{-1}\mathbf{1}\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}}{\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1}}, \tag{39}$$

exactly the projection of Eq. (12).

The same computation has a standard matrix name. Expanding the exponent of Eq. (37) in the joint variables (β, B) gives a Gaussian with the $(n+1)\times(n+1)$ block precision

$$\mathsf{F}_{(\beta,B)} \;=\; \begin{pmatrix} \mathsf{C}^{-1} & -\,\mathsf{C}^{-1}\mathbf{1} \\ -\,\mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1} & \mathbf{1}^{\mathsf{T}}\mathsf{C}^{-1}\mathbf{1} \end{pmatrix}. \tag{40}$$

Removing a parameter from a precision matrix is not just deleting its row and column: the remaining block must also be corrected downward by whatever the removed parameter could absorb. The corrected block, $\mathsf{P} = \mathsf{F}_{\beta\beta} - \mathsf{F}_{\beta B}\,\mathsf{F}_{BB}^{-1}\,\mathsf{F}_{B\beta}$, is called the *Schur complement* over B, and it reproduces Eq. (12) term by term. So P is not a trick: it is precisely the statement "*B_g is sampled with a flat prior and marginalized*", folded into one matrix.

Note also that B_* in Eq. (38) is the same generalized-least-squares mean as a in Eq. (31): the marginalization integrates over all amplitudes, and the best-fitting one dominates the integral.

B.4 Checking views (b) and (c) on the two-bin case

For $n = 2$ the rows of C^{-1} in Eq. (24) have equal sums, so Eq. (31) gives the plain mean, $a = (\Delta\beta_1 + \Delta\beta_2)/2$, and the residual is the antisymmetric half-difference, $w = \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)\,(1,-1)^{\mathsf{T}}$. Its penalty, using the shape-change result of Eq. (26) with $a \to \frac{1}{2}(\Delta\beta_1 - \Delta\beta_2)$, is

$$w^{\mathsf{T}}\mathsf{C}^{-1}w \;=\; \frac{2}{s^2(1-r)}\left(\frac{\Delta\beta_1 - \Delta\beta_2}{2}\right)^2 \;=\; \frac{(\Delta\beta_1 - \Delta\beta_2)^2}{2s^2(1-r)}, \tag{41}$$

which is the direct two-bin computation of Eq. (27). Views (a), (b) and (c) all land on the same matrix.

C Many bins and several groups

C.1 Three bins: total correlation versus direct links

A group now has n bins at redshifts $z_1 < z_2 < \cdots < z_n$, unevenly spaced in general, with the prior covariance of Eq. (5),

$$\mathsf{C}_{ij} \;=\; s^2 \exp\!\left(-\frac{|z_i - z_j|}{L}\right). \tag{42}$$

This matrix is positive definite for any set of distinct redshifts (all its eigenvalues are positive), so the prior is a proper probability distribution and C^{-1} exists.

The structure of C^{-1} is best seen on three evenly spaced bins. With neighbor correlation $r = e^{-\Delta z/L}$, the outer pair sits two steps apart and carries $e^{-2\Delta z/L} = r^2$, so

$$\mathsf{C} \;=\; s^2 \begin{pmatrix} 1 & r & r^2 \\ r & 1 & r \\ r^2 & r & 1 \end{pmatrix}, \qquad \mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r & 0 \\ -r & 1+r^2 & -r \\ 0 & -r & 1 \end{pmatrix}. \tag{43}$$

One multiplication confirms the surprising entry. The $(1,3)$ entry of $\mathsf{C}^{-1}\mathsf{C}$ must be zero (it is an off-diagonal entry of the identity matrix), and indeed row 1 of C^{-1} times column 3 of C gives, dropping the prefactors,

$$1 \cdot r^2 \;+\; (-r)\cdot r \;+\; 0 \cdot 1 \;=\; r^2 - r^2 \;=\; 0. \tag{44}$$

Here is the point that looks paradoxical at first. Bins 1 and 3 *are* correlated (C contains r^2), yet their entry in C^{-1} is exactly zero. The two matrices answer different questions. C gives the *total* correlation: what δ_1 tells you about δ_3 when nothing else is known. C^{-1} gives the *direct* correlation: what δ_1 still tells you about δ_3 once δ_2 is already known, and that is nothing. All of the 1–3 correlation travels through bin 2.

Think of a chain of people passing a message. The first person's message reaches far down the chain, but the links are only between neighbors: nobody talks past the person standing next to them.

The exponential kernel is special in making this exact, because its correlations multiply along the chain:

$$e^{-(z_3 - z_1)/L} \;=\; e^{-(z_3 - z_2)/L}\, e^{-(z_2 - z_1)/L}. \tag{45}$$

The long-distance correlation is the product of the step correlations, so the middle bin already carries everything the outer bins share. A Gaussian kernel $e^{-\Delta z^2/L^2}$ does not factorize this way, because the squared separations do not add up along the chain: $(z_3 - z_1)^2 \neq (z_3 - z_2)^2 + (z_2 - z_1)^2$. Its inverse is dense.

The same structure shows directly in the penalty. For n evenly spaced bins, the quadratic form of the deviations δ_i (whose covariance is $\sigma_b^2 \mathsf{R}$) regroups into a sum over neighbor links,

$$\delta^{\mathsf{T}} \big(\sigma_b^2 \mathsf{R}\big)^{-1} \delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r\,\delta_k\big)^2}{\sigma_b^2\,(1-r^2)}. \tag{46}$$

Each term compares one bin with r times the bin before it: every bin is pulled only by its neighbor, and the long-range correlation comes from chaining the pulls. (Expanding the squares and collecting terms reproduces Eq. (43) for $n = 3$; Appendix D does it for general, uneven spacing.) This is the precise sense of "correlation only between adjacent bins": *directly*, only neighbors talk; *in total*, the influence propagates down the whole chain. Probability theory calls this neighbors-only structure the Markov property.

C.2 The tridiagonal inverse for n bins

A matrix whose nonzero entries sit only on the diagonal and the two neighboring off-diagonals is called *tridiagonal*. The three-bin inverse of Eq. (43) is the smallest interesting example, and the pattern grows unchanged. For n evenly spaced bins, where every adjacent pair has the same correlation $r = e^{-\Delta z/L}$ and $\mathsf{C}_{ij} = s^2\, r^{|i-j|}$,

$$\mathsf{C}^{-1} \;=\; \frac{1}{s^2\,(1-r^2)} \begin{pmatrix} 1 & -r & & & \\ -r & 1+r^2 & -r & & \\ & -r & \ddots & \ddots & \\ & & \ddots & 1+r^2 & -r \\ & & & -r & 1 \end{pmatrix}, \tag{47}$$

with diagonal $(1,\ 1+r^2,\ \ldots,\ 1+r^2,\ 1)$, off-diagonal entries $-r$, blanks zero, and the overall factor $1/\big[s^2(1-r^2)\big]$. The entries read off the chain sum of Eq. (46). An off-diagonal entry is the link between that pair of neighbors. An interior bin touches two links, one to the bin on its left and one to the bin on its right, and the two pulls together contribute $1+r^2$; the first and last bins touch only one link each, hence the plain 1 in the corners.

With uneven spacing, each adjacent pair has its own step and therefore its own correlation,

$$r_k \;=\; \exp\!\left(-\,\frac{z_{k+1}-z_k}{L}\right), \qquad k = 1, \ldots, n-1, \tag{48}$$

and the inverse keeps the same shape with per-pair entries:

$$s^2\left(\mathsf{C}^{-1}\right)_{kk} = \begin{cases} \dfrac{1}{1-r_1^2}, & k=1, \\[2ex] \dfrac{1}{1-r_{k-1}^2} + \dfrac{r_k^2}{1-r_k^2}, & 1<k<n, \\[2ex] \dfrac{1}{1-r_{n-1}^2}, & k=n, \end{cases} \qquad s^2\left(\mathsf{C}^{-1}\right)_{k,k+1} = \frac{-\,r_k}{1-r_k^2}. \tag{49}$$

In words: the off-diagonal entry is the neighbor link of pair k, stronger when the bins are closer (r_k nearer one); the diagonal entry of bin k collects the links touching that bin, one term from the pair on its left and one from the pair on its right; the first and last bins carry one link only. Setting all $r_k = r$ turns the interior diagonal into $1/(1-r^2) + r^2/(1-r^2) = (1+r^2)/(1-r^2)$ and recovers Eq. (47). Appendix D derives Eq. (49) from the chain of neighbor links; the companion script `verify_prior.py` checks the three-bin, even and uneven forms numerically.

C.3 The projection, group by group

Each group receives the projected precision of Eq. (12), built from its own C. The properties derived in Appendix B carry over verbatim to n bins. $\mathsf{P}\,\mathbf{1} = 0$: a uniform shift of the whole group costs nothing. P is positive semidefinite, which means no displacement is *rewarded*: every move either costs χ^2 or is free. And of the n directions a group can move in, exactly one (the uniform shift) is free while the other $n-1$ are charged: in matrix language, P has rank $n-1$.

Several groups combine trivially. The prior of Eq. (3) is zero across groups, so the full prior covariance is block-diagonal, one block per group; and each group has its own free amplitude B_g, so each block is projected separately. The total prior precision is the block-diagonal matrix of the per-group projections.

C.4 The two limits

The intermediate model contains both extremes.

As $L \to \infty$, every $r_k \to 1$, so the jump variance $2\sigma_b^2(1 - r_k)$ of every adjacent pair goes to zero: the prior forces all deviations of a group to be equal. The only freedom left is their common value, a uniform shift, and that is the direction the projection leaves free. This is the tied model, with one free B_g per group, *whatever* the value of σ_b.

As $\sigma_b \to \infty$ at fixed L, the overall scale of C diverges, so $\mathsf{C}^{-1} \to 0$ and with it $\mathsf{P} \to 0$: the prior penalty vanishes and every bin's amplitude is free: the independent model with flat priors.

Between the limits, L and σ_b interpolate continuously, which is what makes the model useful: the figure of merit can be traced as a function of how much bin-to-bin freedom the bias is granted.

D The tridiagonal inverse from the chain of links

Because correlations multiply along the chain (Eq. (45)), the prior can be built one bin at a time: the joint density of the deviations factorizes into neighbor conditionals,

$$p(\delta_1, \ldots, \delta_n) \;=\; p(\delta_1)\, p(\delta_2 \mid \delta_1) \cdots p(\delta_n \mid \delta_{n-1}), \tag{50}$$

where each conditional is Gaussian with

$$\big\langle \delta_{k+1} \mid \delta_k \big\rangle \;=\; r_k\, \delta_k, \qquad \mathrm{Var}\big(\delta_{k+1} \mid \delta_k\big) \;=\; \sigma_b^2\, \big(1 - r_k^2\big). \tag{51}$$

Here $r_k = e^{-(z_{k+1} - z_k)/L}$ is, as in Sections 2.2 and C.2, the correlation coefficient of the adjacent bins k and $k{+}1$: the kernel of Eq. (3) gives $\mathrm{Cov}(\delta_k, \delta_{k+1}) = \sigma_b^2\, r_k$ and $\mathrm{Var}(\delta_k) = \mathrm{Var}(\delta_{k+1}) = \sigma_b^2$.

The conditional mean and variance in Eq. (51) follow from nothing more than these covariances. Split δ_{k+1} into the part predicted by its neighbor and a leftover e,

$$\delta_{k+1} \;=\; a\, \delta_k + e, \tag{52}$$

and choose the number a so that the leftover carries no correlation with δ_k:

$$\mathrm{Cov}(e, \delta_k) \;=\; \mathrm{Cov}(\delta_{k+1}, \delta_k) - a\, \mathrm{Var}(\delta_k) \;=\; \sigma_b^2\, r_k - a\, \sigma_b^2 \;=\; 0 \quad\Longrightarrow\quad a = r_k. \tag{53}$$

The leftover's variance follows by expanding $e = \delta_{k+1} - r_k \delta_k$:

$$\mathrm{Var}(e) \;=\; \mathrm{Var}(\delta_{k+1}) - 2 r_k\, \mathrm{Cov}(\delta_{k+1}, \delta_k) + r_k^2\, \mathrm{Var}(\delta_k) \;=\; \sigma_b^2 - 2 r_k^2 \sigma_b^2 + r_k^2 \sigma_b^2 \;=\; \sigma_b^2\, \big(1 - r_k^2\big). \tag{54}$$

For jointly Gaussian variables, uncorrelated means independent. So once δ_k is known, the only randomness left in $\delta_{k+1} = r_k \delta_k + e$ is e itself: the conditional mean is $r_k \delta_k$ and the conditional variance is $\sigma_b^2(1 - r_k^2)$, which is Eq. (51). In one line: knowing the neighbor explains the fraction r_k^2 of the bin's variance, and the fraction $1 - r_k^2$ is new.

The factorization of Eq. (50) needs one more fact: conditioning on δ_k alone must be enough: the earlier bins must add nothing. This is again the multiplication of correlations along the chain. For any earlier bin $j \le k$,

$$\mathrm{Cov}(e, \delta_j) \;=\; \sigma_b^2 \left[e^{-(z_{k+1}-z_j)/L} - r_k\, e^{-(z_k-z_j)/L} \right] \;=\; \sigma_b^2\, e^{-(z_k-z_j)/L} \left[e^{-(z_{k+1}-z_k)/L} - r_k \right] \;=\; 0, \tag{55}$$

because the first exponential factorizes through z_k and the bracket is $r_k - r_k$. The leftover is therefore independent of every earlier bin, and the chain of one-step conditionals is the complete prior.

Taking $-2\ln p$ of Eq. (50) gives the penalty as the chain sum, the uneven-spacing version of Eq. (46):

$$\delta^{\mathsf{T}} \big(\sigma_b^2 \mathsf{R}\big)^{-1} \delta \;=\; \frac{\delta_1^2}{\sigma_b^2} \;+\; \sum_{k=1}^{n-1} \frac{\big(\delta_{k+1} - r_k\, \delta_k\big)^2}{\sigma_b^2 \left(1 - r_k^2\right)}. \tag{56}$$

Expanding each square and collecting the coefficient of every product $\delta_i \delta_j$ reads off the precision matrix entry by entry. δ_1^2 appears twice: in the first term (coefficient 1) and in the $k = 1$ link as the predictor (coefficient $r_1^2/(1-r_1^2)$); the two add up to $1/(1-r_1^2)$. An interior δ_k^2 appears in the $k-1$ link as the predicted bin ($1/(1-r_{k-1}^2)$) and in the k link as the predictor ($r_k^2/(1-r_k^2)$). The last bin is only ever predicted: $1/(1-r_{n-1}^2)$. Each product $\delta_k \delta_{k+1}$ appears in its own link only, with coefficient $-2r_k/(1-r_k^2)$, i.e. a symmetric matrix entry $-r_k/(1-r_k^2)$. No product of non-neighbors ever appears, so the matrix is tridiagonal by construction. This is Eq. (49) in units of σ_b^2; replacing δ by $\Delta\beta = B_g\, \delta$ and σ_b by $s = \sigma_b B_g$ gives the same matrix for the bin amplitudes.

E The memory projection, step by step

Throughout, a group has n bins, C is the kernel covariance $\mathsf{C}_{ij} = \sigma_b^2 e^{-|z_i - z_j|/L}$ of the residuals, $\mathsf{Q} = \mathsf{C}^{-1}$, $\mathsf{U} = [\,\mathbf{1},\ v\,]$ is the $n \times 2$ design matrix of Eq. (21), and $\theta = (a, s_g)^{\mathsf{T}}$ collects a line's level and slope. (For the bin amplitudes β, every covariance carries a factor B_g^2 that cancels in each identity below.)

E.1 The line fit, the split, and the marginalization

The three views of Appendix B generalize from the single column $\mathbf{1}$ to U without change of structure.

Minimizing the penalty of a displacement δ over the line, $\chi^2(\theta) = (\delta - \mathsf{U}\theta)^{\mathsf{T}} \mathsf{Q}\, (\delta - \mathsf{U}\theta)$, gives the normal equations of a generalized least-squares fit,

$$\mathsf{U}^{\mathsf{T}} \mathsf{Q}\, \mathsf{U}\, \hat{\theta} \;=\; \mathsf{U}^{\mathsf{T}} \mathsf{Q}\, \delta, \qquad \hat{\theta} = (\hat{a}, \hat{s}_g)^{\mathsf{T}}, \tag{57}$$

with $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U}$ a 2×2 matrix, positive definite whenever the group has at least two distinct redshifts (then $\mathbf{1}$ and v are linearly independent and U has full column rank).

Writing $\delta = \mathsf{U}\hat{\theta} + w$, the normal equations say $\mathsf{U}^{\mathsf{T}}\mathsf{Q}\, w = 0$: the residual is orthogonal to both columns in the Q metric. The cross terms then vanish and the penalty splits,

$$\delta^{\mathsf{T}} \mathsf{Q}\, \delta \;=\; \hat{\theta}^{\mathsf{T}} \big(\mathsf{U}^{\mathsf{T}} \mathsf{Q}\, \mathsf{U}\big) \hat{\theta} \;+\; w^{\mathsf{T}} \mathsf{Q}\, w, \qquad \delta^{\mathsf{T}} \mathsf{P}_{\mathrm{mem}}\, \delta \;=\; w^{\mathsf{T}} \mathsf{Q}\, w : \tag{58}$$

the projection keeps the residual's penalty and discards the line's. A pure line has $w = 0$, so $\mathsf{P}_{\rm mem}\mathsf{U} = 0$; the residuals obey two linear constraints, so $\mathsf{P}_{\rm mem}$ has rank $n-2$; and $w^{\mathsf{T}}\mathsf{Q}\,w \geq 0$ makes it positive semidefinite.

Marginalization gives the same matrix. With $\delta \sim \mathcal{N}(\mathsf{U}\theta, \mathsf{C})$ and $\theta \sim \mathcal{N}(0, \Lambda)$, $\Lambda = \mathrm{diag}(W^2, \tau^2)$, the marginal covariance of δ is $\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}$, and the Woodbury identity inverts it:

$$\big(\mathsf{C} + \mathsf{U}\Lambda\mathsf{U}^{\mathsf{T}}\big)^{-1} \;=\; \mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\Lambda^{-1} + \mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q}. \tag{59}$$

As the prior widths grow, $\Lambda^{-1} \to 0$ and the right-hand side becomes $\mathsf{P}_{\rm mem}$. The deviation falls inverse-quadratically in the widths: expanding the middle inverse, the leading correction carries Λ^{-1}, so the deviation falls as $1/W^2$, a factor of one hundred per decade of W. The script of Appendix I verifies the fall.

E.2 The null-space form

A second form of the projection makes its meaning plain. Let N be an $n \times (n-2)$ matrix whose columns span the combinations of bins that no straight line can produce: $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$, plain Euclidean orthogonality to $\mathbf{1}$ and to v. Then

$$\mathsf{Q} - \mathsf{Q}\,\mathsf{U}\big(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\,\mathsf{U}\big)^{-1}\mathsf{U}^{\mathsf{T}}\mathsf{Q} \;=\; \mathsf{N}\big(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N}\big)^{-1}\mathsf{N}^{\mathsf{T}}. \tag{60}$$

The proof has three steps. Both sides annihilate U: the left by construction, the right because $\mathsf{N}^{\mathsf{T}}\mathsf{U} = 0$. Both sides map CN to N: the right because $\mathsf{N}(\mathsf{N}^{\mathsf{T}}\mathsf{CN})^{-1}(\mathsf{N}^{\mathsf{T}}\mathsf{CN}) = \mathsf{N}$, the left because $\mathsf{QCN} = \mathsf{N}$ and the subtracted term meets $\mathsf{U}^{\mathsf{T}}\mathsf{N} = 0$. And the columns of CN and of U together span the whole space: if $\mathsf{CN}x + \mathsf{U}y = 0$, multiplying by N^{T} gives $\mathsf{N}^{\mathsf{T}}\mathsf{CN}\,x = 0$, so $x = 0$ (the middle matrix is positive definite), and then $y = 0$ (U has full column rank). Two matrices that agree on a spanning set are equal.

The meaning: since $\delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta = (\mathsf{N}^{\mathsf{T}}\delta)^{\mathsf{T}}(\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N})^{-1}(\mathsf{N}^{\mathsf{T}}\delta)$, the prior reads off the combinations $\mathsf{N}^{\mathsf{T}}\delta$ that no line can make and charges each by the inverse of its own prior covariance. The same identity with the single column $\mathsf{U} = \mathbf{1}$ gives the kernel-model projection of Eq. (12); there N spans the zero-sum combinations, the differences between bins.

E.3 Worked examples

One or two bins are fully free. For $n = 2$ the matrix U is square and invertible (two distinct redshifts), so $(\mathsf{U}^{\mathsf{T}}\mathsf{Q}\mathsf{U})^{-1} = \mathsf{U}^{-1}\mathsf{C}\,(\mathsf{U}^{\mathsf{T}})^{-1}$ and the subtracted term collapses to Q: the projection vanishes. For $n = 1$ nothing is orthogonal to both columns, so N has no columns and the null-space form is the zero matrix: a line fixes a single value trivially.

Three evenly spaced bins show what survives. With spacing Δ and $r = e^{-\Delta/L}$, the centered redshifts are $v = (-\Delta, 0, \Delta)^{\mathsf{T}}$, and the only direction orthogonal to $\mathbf{1}$ and v is the discrete curvature, the second difference $\mathsf{N} = (1, -2, 1)^{\mathsf{T}}$. Its prior variance expands to

$$\mathsf{N}^{\mathsf{T}}\mathsf{C}\,\mathsf{N} \;=\; \sigma_b^2\,\big(6 - 8r + 2r^2\big) \;=\; 2\,\sigma_b^2\,(1-r)(3-r) \tag{61}$$

(the diagonal contributes $1+4+1 = 6$, the two nearest-neighbor pairs $-8r$, and the outer pair $+2r^2$), so the penalty is

$$\delta^{\mathsf{T}}\mathsf{P}_{\rm mem}\,\delta \;=\; \frac{\big(\delta_1 - 2\delta_2 + \delta_3\big)^2}{2\,\sigma_b^2\,(1-r)(3-r)}. \tag{62}$$

The memory model charges only the curvature; it is blind to any straight-line drift. The kernel model's three-bin case (Appendix C.1) charges the two neighbor differences instead, so it taxes a drift as heavily as a zigzag of the same step size.

E.4 The universal-kriging predictor

Predict bin $k+1$ from the first k bins, $k \geq 2$, with flat priors on θ. Write C_k for the past kernel block, U_k for the past rows of U, $u_{k+1} = (1, v_{k+1})^{\mathsf{T}}$ for the next row, and c for the vector of kernel covariances between bin $k+1$ and the past. Given θ, the ordinary Gaussian conditional is

$$\mathrm{E}\big[\delta_{k+1} \mid \delta_{1..k}, \theta\big] \;=\; u_{k+1}^{\mathsf{T}}\theta \;+\; c^{\mathsf{T}}\mathsf{C}_k^{-1}\big(\delta_{1..k} - \mathsf{U}_k\,\theta\big). \tag{63}$$

With a flat prior, the posterior of θ given the past is centered on the generalized-least-squares fit $\hat{\theta}$ of Eq. (57) applied to the past block. Equation (63) is linear in θ, so averaging over the posterior replaces θ by $\hat{\theta}$.

For the exponential kernel the weights collapse. Correlations multiply along the chain (Eq. (45)), so every entry of c is r_k times the corresponding entry of C_k's last column, and

$$c^{\mathsf{T}}\mathsf{C}_k^{-1} \;=\; r_k\, e_k^{\mathsf{T}}, \tag{64}$$

with e_k the k-th unit vector (one in the last past slot, zero elsewhere). Substituting both results into Eq. (63) gives Eq. (20): the fitted line evaluated at the new bin, plus one Markov step from the last residual.

E.5 The finite-slope correlation

With a Gaussian slope of variance τ^2 and the amplitude held in B_g, the deviations have the covariance of Eq. (18), so $\mathrm{Var}(\delta_i) = \tau^2 v_i^2 + \sigma_b^2$ and $\mathrm{Cov}(\delta_i, \delta_j) = \tau^2 v_i v_j + \sigma_b^2 e^{-|z_i - z_j|/L}$. For separations much larger than L the kernel term is exponentially small, and dropping it gives the correlation coefficient of Eq. (19). When $\tau|v_i|$ and $\tau|v_j|$ both exceed σ_b, the σ_b^2 terms in the denominators become negligible and the coefficient tends to $v_i v_j/|v_i v_j| = \pm 1$: plus one for bins on the same side of the group center, minus one for opposite sides.

F The bias continuity equation, step by step

The matter density $\rho_m(\mathbf{x}, t)$ defines the matter overdensity $\delta_m = (\rho_m - \bar{\rho}_m)/\bar{\rho}_m$, with $\bar{\rho}_m(t)$ the mean density; the galaxy number density n_g defines the galaxy-count overdensity δ_g the same way. Conservation of mass and of galaxy number gives each field a continuity equation. In comoving coordinates, keeping only terms linear in the fluctuations (the convective term $\mathbf{v} \cdot \nabla\delta$ is second order and dropped),

$$\frac{\partial \delta_m}{\partial t} \;=\; -\frac{1}{a}\,\nabla \cdot \mathbf{v}_m, \qquad \frac{\partial \delta_g}{\partial t} \;=\; -\frac{1}{a}\,\nabla \cdot \mathbf{v}_g, \tag{65}$$

where t is cosmic time, a the scale factor, and $\mathbf{v}_m$ and $\mathbf{v}_g$ the peculiar velocity fields of the matter and of the galaxies.

The two velocity fields are the same. On large scales the galaxies carry no pressure of their own and fall in the same gravitational potential as the matter, so a conserved sample rides the

matter flow: $\mathbf{v}_g = \mathbf{v}_m$. This is the no-velocity-bias assumption; Tegmark and Peebles [2] state it as galaxies behaving as test particles that fairly trace the large-scale velocity field.

With equal right-hand sides, subtracting the two equations of Eq. (65) leaves

$$\frac{\partial}{\partial t}\big(\delta_g - \delta_m\big) \;=\; 0 \quad\Longrightarrow\quad \delta_g(z) - \delta_m(z) \;=\; \delta_g(z_c) - \delta_m(z_c) \tag{66}$$

after integrating in time from the reference redshift z_c to any other redshift z: the excess clumping keeps its initial value forever.

Now substitute the bias. At z_c the sample has linear bias b_c, so $\delta_g(z_c) = b_c\,\delta_m(z_c)$ and the frozen difference is $(b_c - 1)\,\delta_m(z_c)$. The matter grows as $\delta_m(z) = \delta_m(z_c)\,D(z)/D(z_c)$, with D the linear growth factor. Then

$$\delta_g(z) \;=\; \delta_m(z) + \big(b_c - 1\big)\,\delta_m(z_c) \;=\; \delta_m(z)\left[\,1 + \big(b_c - 1\big)\,\frac{D(z_c)}{D(z)}\,\right], \tag{67}$$

and dividing by $\delta_m(z)$ gives the bias $b(z) = \delta_g/\delta_m$ of Eq. (13). Dividing instead by the tied bias $b_c\,D(z_c)/D(z)$ gives the deviation of Eq. (14), $\delta(z) = (D(z)/D(z_c) - 1)/b_c$: positive at redshifts below z_c, where the matter has grown past its reference value, and negative above.

The debiasing limit follows from Eq. (13) alone. As D grows, the frozen term is divided by an ever larger number, so $b \to 1$: from above when $b_c > 1$ and from below when $b_c < 1$. Gravity debiases a conserved sample, because the part of δ_g that traces the matter keeps growing while the frozen excess does not.

Tegmark and Peebles drop the assumption that δ_g is proportional to δ_m point by point. They describe the pair of fields by three numbers: the matter fluctuation amplitude σ, the bias b (the ratio of the galaxy to the matter fluctuation amplitude) and the correlation coefficient ρ_{gm} between the two fields; $\rho_{gm} < 1$ is stochastic bias, the galaxy field tracing the matter imperfectly. (The subscript keeps it apart from the neighbor correlation r of Eq. (7).) The same continuity argument evolves all three (their section "After the galaxy formation epoch"): with D normalized to one today and the subscript zero marking today,

$$b_0 = \big[(1-D)^2 + 2D(1-D)\,b\,\rho_{gm} + D^2 b^2\big]^{1/2}, \qquad \rho_{gm,0} = \frac{(1-D) + D\,b\,\rho_{gm}}{b_0}, \tag{68}$$

where b and ρ_{gm} are the values at the epoch whose growth factor is D.

Both b_0 and $\rho_{gm,0}$ approach one as clustering grows, whatever their initial values: the matter-sourced part of the galaxy field grows with D while the frozen part does not, so the field is progressively dominated by the piece that traces the matter. Perfect correlation is a fixed point: for $\rho_{gm} = 1$, Eq. (68) collapses to $b_0 - 1 = (b-1)\,D$, which is Eq. (13) again. The conserved population of this note is the perfectly correlated special case of their model.

Each assumption fails in a specific way for a DESI sample. Membership is not conserved: the flux limit and color cuts admit and remove galaxies as redshift grows (the threshold drift and color-selection features of Section 3.4), mergers turn two members into one, and an ELG belongs to the sample only during its star-forming phase. The no-velocity-bias and linear-theory assumptions hold well on the scales this forecast keeps; conservation is the assumption that breaks.

The script of Appendix I integrates Eq. (65) numerically, with δ_m from the linear growth equation, and recovers the closed form of Eq. (13) to a relative accuracy of 2×10^{-9}.

G What the literature says, paper by paper

This appendix summarizes each measurement behind Section 3.5, for a reader who has not read the papers. The tags are those of that section: measured, derived, assumed.

G.1 Passive-evolution tests of massive red galaxies

Wake et al. [3]. What was measured: the two-point correlation function of over 10,000 luminous red galaxies (LRGs) from SDSS and the 2SLAQ survey at $0.19 < z < 0.55$, matched in rest-frame color and luminosity, with stellar masses above $10^{11}\,h^{-1}M_{\odot}$. Method: the redshift-space and real-space autocorrelation of the matched samples, interpreted with a halo model. Result: no significant evolution of the correlation amplitude, but a strictly merger-free passive model is rejected; a merger rate of $7.5 \pm 2.3\%$ between $z = 0.55$ and $z = 0.19$, about 2.4% per Gyr, fits the data. Implication here: LRG deviations from conserved evolution are a few percent in number per Gyr (measured), and the induced bias change is smaller, because merging moves galaxies between halos of similar mass.

Tojeiro et al. [4]. What was measured: the merger rate and clustering evolution of 250,000 BOSS galaxies up to $z \approx 0.7$, connected to the SDSS-I/II LRGs at lower redshift. Method: each BOSS galaxy receives a likelihood of being a progenitor of the LRG sample, built from the fossil record of the LRGs (star-formation and metallicity histories); the weighted sample's large-scale clustering is compared between the surveys. Result: the sample evolves by merging at $2 \pm 1.5\%$ per Gyr, and the weighted clustering evolution is consistent with dynamical passive evolution, the continuity-equation prediction. Implication here: the conserved shape holds to the percent-per-Gyr level for these galaxies (measured).

Guo et al. [5]. What was measured: the projected two-point correlation function of about 260,000 CMASS galaxies at $0.43 < z < 0.7$ (BOSS DR9). Method: subsamples of fixed comoving number density built with redshift-dependent magnitude thresholds, so that the same population is followed across redshift. Result: the clustering of the fixed-density samples is consistent with passive evolution, and red-sequence clustering at fixed luminosity shows no significant evolution. Implication here: the same (measured).

G.2 The DESI luminous red galaxies

Zhou et al. [6]. What was measured: the redshift-dependent clustering of 2.7 million DESI-like LRGs over 5655 square degrees at $0.4 < z < 0.9$, in photometric-redshift bins of width $\Delta z \approx 0.1$. Method: projected correlation functions with photometric redshifts of 0.02 precision, fit with a five-parameter halo occupation distribution (HOD). Result: the HOD parameters show little evolution except in the highest bin, and the bias follows a constant clustering amplitude; Table 5 lists the measured biases and their residuals around $b = 1.5/D$ (derived here): within $\pm 2.3\%$ below $z = 0.83$, and $+6.4 \pm 6.4\%$ at $0.84 < z < 0.93$. Implication: LRG deviations at $\Delta z \approx 0.1$ spacing are at most a few percent below $z = 0.83$ (measured plus derived).

photometric bin	b	residual around $1.5/D$
0.41–0.50	1.89 ± 0.03	$-0.4\% \pm 1.6\%$
0.50–0.61	2.04 ± 0.02	$+2.3\% \pm 1.0\%$
0.61–0.72	2.07 ± 0.03	$-1.6\% \pm 1.4\%$
0.75–0.83	$2.24^{+0.07}_{-0.08}$	$+0.4\% \pm 3.4\%$
0.84–0.93	2.48 ± 0.15	$+6.4\% \pm 6.4\%$

Table 5: The DESI-like LRG bias of Zhou et al. (2021, arXiv:2001.06018, Results section; measured) and its residual around the constant clustering amplitude $b = 1.5/D$ (derived, with flat ΛCDM $\Omega_m = 0.3$ growth factors at the bin centers).

Zhou et al. [7]. What was measured: the DESI LRG target selection and its spectroscopic performance. Method: selection cuts in g, r, z and $W1$ (the WISE 3.4 micron band), validated on Survey Validation spectra. Result: a sliding color-magnitude cut holds the comoving density near $5 \times 10^{-4}\,(h/\mathrm{Mpc})^3$ at $0.4 < z < 0.8$; the overall faint limit is $z_{\mathrm{fiber}} < 21.6$, where z_{fiber} is the z-band magnitude expected within one DESI fiber; 98.9% of observed targets yield confident redshifts. Implication: inside LRG1 and LRG2 the selection gives the threshold no room to jump (measured); past $z \approx 0.8$ the fixed flux limit drives the LRG3 drift of Section 3.5 (derived). The chain of that derivation: a fixed flux limit raises the luminosity threshold with the distance modulus, by 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$; the Zehavi slope below is 0.5–1.0 in bias per magnitude near LRG luminosities; their product, 0.13–0.27 in bias, is 6–11% of $b \approx 2.4$. The caveat is stronger than for BGS: the threshold-bias fit is an r-band relation measured at $z < 0.25$, applied here to a z-band fiber-magnitude selection at $z \approx 0.8$–1.1, outside its calibrated range, so the drift's direction and order of magnitude are what the argument establishes.

White et al. [9] and Sailer et al. [10]. What was measured: cosmological constraints from the cross-correlation of the photometric LRG sample of [8] with CMB lensing from Planck, and in [10] also ACT DR6, in four tomographic bins over $0.4 < z < 1$. Method: angular cross- and auto-spectra with one free bias parameter per bin. Result: smooth models fit all bins. Implication here: these analyses neither require nor exclude percent-level bias structure between thin bins; they bound nothing below their bin width of about 0.15 (measured context).

G.3 Bias versus luminosity threshold

Zehavi et al. [12]. What was measured: the luminosity and color dependence of galaxy clustering in the SDSS DR7 main sample, about 700,000 galaxies at $z < 0.25$. Method: projected correlation functions of volume-limited luminosity-threshold samples, interpreted with HOD modeling. Result: the large-scale bias of a threshold sample is fit by $b\,(\sigma_8/0.8) = 1.06 + 0.21\,(L/L_*)^{1.12}$, with L_* the characteristic luminosity of the r-band luminosity function and σ_8 the rms matter fluctuation in spheres of $8\,\mathrm{Mpc}/h$; the measured threshold biases rise from 1.40 ± 0.03 at $M_r < -21.0$ to 2.16 ± 0.05 at $M_r < -22.0$, a slope of 0.5–1.0 in bias per magnitude of threshold near LRG luminosities. Implication: this relation converts a selection-threshold drift into a bias drift; the LRG3 and BGS drifts of Section 3.5 are derived through it.

G.4 The bright galaxy survey

Hahn et al. [13]. What was measured: the BGS target selection and its validation. Method: selection on Legacy Surveys DR9 imaging plus Survey Validation spectra. Result: BGS Bright is a magnitude-limited sample at $r < 19.5$ with redshift success above 95% across a broad range of galaxies; BGS Faint adds $19.5 < r < 20.175$ with a color selection. Implication: the fixed $r < 19.5$ limit raises the luminosity threshold with redshift, which drives the derived 10–25% bias drift across the BGS group (measured limit, derived drift).

Sailer et al. [11]. What was measured: the growth of matter fluctuations at $z \lesssim 0.4$ from BGS imaging cross-correlated with Planck and ACT CMB lensing. Method: tomographic angular cross-spectra with a hybrid effective-field-theory model and per-bin bias nuisance parameters. Result: growth consistent with ΛCDM. Implication here: as with the LRG tomography, no resolution of bias structure between thin bins (measured context).

G.5 The emission-line galaxies

Raichoor et al. [14]. What was measured: the ELG target selection, $0.6 < z < 1.6$, and its spectroscopic efficiency. Method: a g-band fiber magnitude cut plus a $(g-r)$ versus $(r-z)$ color box; redshift reliability is a cut on the signal-to-noise of the [OII] 3727 Å doublet. Result: the [OII] signal-to-noise drops at the specific redshifts where the doublet lands on sky emission lines, especially at $z > 1.5$, and the catastrophic-failure fraction rises over $1.3 < z < 1.6$. Implication: these windows, each $\Delta z \approx 0.001$–0.01 wide, are the one short-separation selection mechanism found in any tracer; their bias amplitude is assumed, of order 1–2%, a judgment from the weak dependence of ELG bias on [OII] flux rather than a measured bound.

Rocher et al. [15]. What was measured: the HOD of the DESI One-Percent ELG sample. Method: fits of AbacusSummit simulation halos to the measured small-scale clustering, in two redshift bins, $0.8 < z < 1.1$ and $1.1 < z < 1.6$. Result: every HOD parameter agrees within one sigma between the bins; the mean host halo mass is $\log_{10} M_h = 11.78^{+0.03}_{-0.04}$ at mean redshift 0.95 and 11.86 ± 0.05 at 1.325. Implication: the ELG bias shape is stable across ELG1 and most of ELG2 (measured); at these host masses the halo bias depends weakly on mass, so the implied drift is at the percent level (derived, with the slope assumed).

H How the prior enters the Fisher matrix

A Gaussian prior adds its precision to the Fisher matrix:

$$F \;=\; F_{\rm data} \;+\; F_{\rm priors}, \tag{69}$$

where $F_{\rm data}$ is the usual data Fisher matrix and $F_{\rm priors}$ holds one block per prior. The projected precision P of each group is added to the rows and columns of that group's bin amplitudes β_i. All other parameters (the cosmological parameters, and the velocity dispersions with their own independent Gaussian priors) are unchanged: P has no entries outside the bias block.

One consequence matters in practice. Bins that lost all their band powers to the scale cut carry no data, so their rows of $F_{\rm data}$ are zero. In the independent model such a parameter is completely unconstrained and must be dropped from the matrix before inversion. With the

correlated prior it stays: P ties the empty bin to its neighbors, which do carry data, so the matrix remains invertible and the empty bin follows the chain.

I Numerical verification

The companion script `verify_prior.py`, in this folder, checks every identity used above: the two-bin closed forms of Eqs. (24) and (27); the three-bin inverse of Eq. (43), including its exactly zero $(1, 3)$ entry; the null direction, positive semidefiniteness and rank $n - 1$ of P on uneven redshifts; the tridiagonal inverses of Eqs. (47) and (49); the Pythagorean identity of Eq. (35) on random displacements; the equality of P with the Schur complement of the joint (β, B) precision of Eq. (40); the decorrelation and variance of the neighbor-prediction leftover of Appendix D; the jump variance of Eq. (8) by Monte Carlo (10^5 draws); the worked example of Section 2.2; the closed-form bias evolution of Eq. (13) against a direct integration of the continuity equation; and the tilt and neighbor-change tables of Section 3. For the memory model of Section 5 it checks, in the same way: the annihilation $\mathsf{P}_{\rm mem}\mathsf{U} = 0$ with rank $n - 2$ on uneven redshifts; the $1/W^2$ approach of the Woodbury inverse of Eq. (59) to the projection; the null-space identity of Eq. (60); the three-bin curvature penalty with its closed-form variance $2\sigma_b^2(1 - r)(3 - r)$; the vanishing of the projection for one and two bins; the universal-kriging predictor of Eq. (20) against a direct Gaussian conditional; the end-to-end correlation of Eq. (19) tending to ± 1; the two straight-line fits of Section 5.4, the conserved-population table and the BGS flux-limit drift; and the LRG3 magnitude-limit rate, 0.27 magnitudes per $\Delta z = 0.1$ at $z \approx 1$ with its 6–11% bias drift. Each check prints a PASS or FAIL line with its measured deviation.

Every figure of the note lives in its own folder under `plots/`, named after the figure. The folder holds the figure's data (`data.json`), one Python script that draws it, and the PDF the note includes, so each figure redraws from its folder alone and a plot can be adjusted for the paper without rerunning the forecast. The script `plots/make_theory_data.py` writes the theory data, from the same flat ΛCDM growth factor and project fiducial amplitudes as the checks above; the notebook `EXAMPLE_EVALUATE_FISHER.ipynb` exports the FoM data.

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